Lesson **40 minutes · concept**

ICT Level 2 — Complete Lesson Plans

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Chapter 1 — Networking and Internet

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Engage (5 min)

Teacher Students, imagine you are in the dormitory and you want to share a photo from your phone to your friend's laptop. How would you do it? Think about all the ways information travels between devices.

Students Students mention Bluetooth, Wi-Fi, cables, and USB drives. They discuss which method is fastest and why.

Explore (10 min)

Teacher Look at the network diagram on the board. You can see three computers connected in the classroom. Now look at the bigger diagram showing connections between our school and other schools in the district. What differences do you notice in the size and distance of these connections?

Students Students observe that classroom network is small and short distance while district network connects many buildings over long distance. They note more devices in larger network.

Explain (10 min)

Teacher A computer network is a group of devices connected together to share data and resources. Networks are classified by size. PAN covers one person like phone to headphones. LAN covers one building like our school. MAN covers a city. WAN covers countries or the whole world like the Internet.

Students Students take notes on network types with examples. They create a comparison table showing range, speed, and examples for PAN, LAN, MAN, and WAN.

Elaborate (10 min)

Teacher Think about KREIS school system. We have multiple schools across Karnataka. What type of network would connect all KREIS schools together? What type of network exists inside each school computer lab?

Students Students identify that inter-school connection is WAN while school lab network is LAN. They draw a simple network diagram showing KREIS school connections.

Evaluate (5 min)

Teacher Match each network type to its correct example. Write one example from your daily life for each network type.

Students Students correctly match PAN to phone headset, LAN to computer lab, MAN to city network, and WAN to Internet. They provide personal examples like smartwatch to phone.

Cross-Curricular Connections

Mathematics: Calculate distances between KREIS schools using map scale. Add distances in kilometers.
Science: Discuss how data travels through cables and air waves. Explain radio frequency used by Wi-Fi.
Language: Write a short paragraph describing how networks help in communication between KREIS hostels.

Digital Citizen Reminder: Remember that every device on a network can potentially access your personal data. Always think before connecting to unknown networks.

Resources Needed:
- Network diagram poster
- Chart paper
- Markers
- Textbook Chapter 1
- Computer with projector

Differentiation

Approaching: Provide pre-made flashcards with network definitions and matching pictures for visual support
On Level: Students complete standard matching and labeling activities from textbook
Advanced: Challenge students to research and present about satellite networks and their use in rural Karnataka
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Worksheet matching network types to examples
CT Skill Assessed: Classification by grouping similar items based on properties

Dormitory — Retrieval (Paper)

Tonight ask your hostel warden what type of network the school uses. Write down three devices connected to the school network.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Close your eyes and imagine two scenarios. First, you are sharing a file with the student sitting next to you in class. Second, you are sending a message to your friend in another KREIS school fifty kilometers away. What is different about these two situations?

Students Students discuss proximity, speed, and complexity. They realize short distance sharing is simpler than long distance communication.

Explore (10 min)

Teacher Examine this comparison table on the board. LAN column has short distance, high speed, private ownership. WAN column has long distance, lower speed, public or rented. Now look at the network map showing our school LAN connecting to the Internet which is the biggest WAN.

Students Students identify that LAN connects devices in one location while WAN connects distant locations. They note LAN is faster and more secure than WAN.

Explain (10 min)

Teacher LAN or Local Area Network connects devices within a small area like a building or campus. It uses Ethernet cables or Wi-Fi. Speed is high from one hundred Mbps to ten Gbps. WAN or Wide Area Network connects devices over large distances using telephone lines, fiber optics, or satellites. Internet is the largest WAN in the world.

Students Students create a Venn diagram comparing LAN and WAN. They list five features of each and identify two features common to both.

Elaborate (10 min)

Teacher Our KREIS residential school has a LAN in the computer lab. When we connect to the Internet we are joining a WAN. Design a simple network for a KREIS school campus showing the computer lab, library, staff room, and hostel all connected.

Students Students sketch a campus LAN with labeled devices. They show the Internet connection as a cloud outside the campus boundary.

Evaluate (5 min)

Teacher Answer these questions. What is the full form of LAN? Give two advantages of LAN over WAN. Name one WAN you use every day. Draw a simple LAN diagram.

Students Students answer correctly showing understanding of definitions, advantages, and real-world examples. Diagrams show proper device connections.

Cross-Curricular Connections

Mathematics: Measure the length of ethernet cable needed to connect two computers in the lab. Calculate cable cost per meter.
Science: Understand how fiber optic cables transmit data using light signals through glass fibers.

Language: Write a comparison paragraph about LAN and WAN using connectives like however, whereas, and on the other hand.

Digital Citizen Reminder: LAN networks in schools are monitored for safety. Respect the school network rules and never try to access restricted areas of the network.

Resources Needed:
- Ethernet cable sample
- LAN diagram handout
- Textbook page 12
- Whiteboard and markers
- Computer lab access

Differentiation

Approaching: Provide sentence starters for comparing LAN and WAN with guided word banks
On Level: Students complete standard comparison chart and answer textbook questions
Advanced: Advanced students research metropolitan area networks and describe how they differ from WAN
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Comparison chart and short answer questions
CT Skill Assessed: Comparative analysis identifying similarities and differences between systems

Dormitory — Retrieval (Paper)

Count how many devices in your dormitory room connect to the school Wi-Fi. Write their names and guess their maximum speed.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

> **Teacher** What do you do when you first open a browser on the school computer? Think about all the different things you can do on the Internet in one hour. List as many activities as you can think of.

Students Students brainstorm activities like watching videos, sending messages, searching for information, downloading files, reading news, and playing games online.

Explore (10 min)

Teacher Look at these screenshots on the board. The first shows a website browser. The second shows an email inbox. The third shows a file download. The fourth shows a video call. For each screenshot, identify what Internet service is being used.

Students Students identify web browsing, email, file transfer, and video conferencing. They note that each service serves a different purpose.

Explain (10 min)

Teacher The Internet provides many services. World Wide Web lets you access websites and information. Email lets you send and receive electronic messages. FTP lets you transfer files between computers. Video conferencing lets you have face-to-face conversations online. Social media lets you connect with friends and share content.

Students Students create a mind map with Internet services at the center branching out to each service with examples. They write one advantage of each service.

Elaborate (10 min)

Teacher Think about a day in KREIS school without Internet services. How would your learning be affected? Which Internet service is most important for your studies and why? Present your reasoning to the class.

Students Students argue for different services based on educational value. Some choose web search for research, others choose email for communication with teachers, and some choose video for learning.

Evaluate (5 min)

Teacher Draw a table with columns Service, Purpose, and Example. Fill in at least five Internet services with their purposes and real examples you have used.

Students Students create accurate tables showing web browsing for information, email for communication, FTP for file sharing, streaming for entertainment, and cloud storage for backup.

Cross-Curricular Connections

Mathematics: Calculate the time saved using email instead of postal mail to send a message from Bidar to Bengaluru.
Science: Explain how data packets travel through Internet infrastructure using radio waves, cables, and satellites.
Language: Write a formal email to a teacher requesting study material using proper email format with subject, greeting, body, and signature.

Digital Citizen Reminder: Not all Internet services are equally safe. Verify the authenticity of websites before entering personal information. Use secure connections for banking and shopping.

Resources Needed:
- Screenshot prints of different services
- Projector

- Textbook
- Worksheet with service table
- Internet access for demo

Differentiation

Approaching: Provide a partially filled table with picture prompts for each Internet service
On Level: Students complete the standard activity identifying and describing five Internet services
Advanced: Challenge students to evaluate and rank Internet services by importance for rural education in Karnataka
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Service identification worksheet with screenshots
CT Skill Assessed: Functional analysis understanding how different tools serve specific purposes

Dormitory — Retrieval (Paper)

List all Internet services you used today before sleeping. Mark each with safe or risky and explain your rating.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

> **Teacher** Imagine your school principal wants to send an important announcement to all KREIS teachers and also wants to copy the district education officer. What email features would help in this situation? Think about who should see the email and who should remain hidden.

> **Students** students discuss CC for visible recipients, BCC for hidden recipients, and the importance of professional email communication in official settings.

Explore (10 min)

> **Teacher** Examine these three emails on the board. The first is informal with no subject and spelling errors. The second is formal with proper greeting and structure. The third has a suspicious link and unknown sender. Which email would you trust and why?

Students students identify that formal structured emails are more professional and trustworthy. They notice phishing signs like suspicious links and unknown senders in the third email.

Explain (10 min)

Teacher Professional emails have a clear subject line, proper greeting like Dear or respected, concise body paragraphs, and a formal closing with your name and class. CC or carbon copy sends the email to additional visible recipients. BCC or blind carbon copy sends the email without other recipients knowing. Attachments let you include files like documents or images.

Students students practice writing a formal email to a librarian requesting book reservation. They use CC to include their class teacher and BCC to inform the warden secretly.

Elaborate (10 min)

Teacher Write an email to the district education officer requesting new computers for the KREIS school lab. Include your class teacher in CC and the principal in BCC. Attach a list of required items as a separate document.

Students students compose complete professional emails with all elements. They verify correct use of CC and BCC and practice attaching files using the school computers.

Evaluate (5 min)

Teacher Review the email you wrote. Check for proper subject line, greeting, body structure, closing, and correct use of CC and BCC. Rate your email on a scale of one to five for professionalism.

Students students self-evaluate their emails against a rubric. They identify areas for improvement and revise their drafts for clarity and professionalism.

Cross-Curricular Connections

Mathematics: Calculate the number of recipients who will receive a BCC email when ten teachers are blind copied on a message to fifty parents.
Science: Understand how email servers use encryption protocols to protect message content during transmission across networks.
Language: Practice formal letter writing skills by converting email format to traditional letter format and compare the differences.

Digital Citizen Reminder: Never share your email password with anyone. Enable two-factor authentication for extra security. Report suspicious emails to your teacher immediately.

Resources Needed:
- Email templates
- Phishing example printouts
- Computer with email client
- Rubric for email assessment
- Textbook

Differentiation

Approaching: Provide email templates with fill-in-the-blank sections for subject, greeting, and closing
On Level: Students write complete emails following the standard professional email format
Advanced: Advanced students analyze real phishing emails and create a detection guide for classmates
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Email writing rubric evaluating structure, tone, and technical accuracy
CT Skill Assessed: Structured communication organizing information according to formal conventions

Dormitory — Retrieval (Paper)

Write a draft email to your parents describing your week at KREIS school. Include at least five sentences with proper greeting and closing.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Look at how the desks are arranged in our classroom. Some are in rows, some are in a circle, and some are grouped together. If each desk represented a computer, how would you connect them? Think about the shape of connections.

Students students suggest different connection patterns like straight lines, circles, and web-like patterns. They realize connection shape affects communication flow.

Explore (10 min)

Teacher Use these yarn pieces and cards to create four different network shapes on the floor. First arrange them in a straight line like a bus. Then arrange them in a circle like a ring. Next arrange them around one central card like a star. Finally connect every card to every other card like a mesh.

Students students physically create topology models. They observe that mesh uses the most yarn and star uses a central hub. Bus is simplest to create.

Explain (10 min)

Teacher Topology is the arrangement of devices in a network. Bus topology connects all devices to one central cable like beads on a string. Star topology connects all devices to one central hub like spokes of a wheel. Ring topology connects devices in a closed loop like a necklace. Mesh topology connects every device to every other device like a spider web.

Students students draw diagrams of all four topologies and label each component. They create a comparison table listing cost, reliability, and troubleshooting difficulty for each.

Elaborate (10 min)

Teacher Our school computer lab uses one of these topologies. Observe the cable connections in the lab and identify the topology used. Explain why this topology was chosen for our lab based on its advantages.

Students students identify star topology as most common in labs because if one cable fails others continue working. They discuss how central switch makes troubleshooting easy.

Evaluate (5 min)

Teacher Draw all four network topologies from memory. Label the components of each. Write two advantages and two disadvantages of star topology compared to bus topology.

Students students produce accurate diagrams with correct labels. They articulate clear comparisons between topologies showing understanding of trade-offs.

Cross-Curricular Connections

Mathematics: Calculate the number of cables needed for mesh topology with five, six, and seven computers. Find the mathematical pattern.
Science: Explain how electrical signals travel differently through bus topology cable compared to star topology hub and cable connections.
Language: Write a persuasive paragraph arguing which topology is best for a school network using evidence and reasoning.

Digital Citizen Reminder: Network topology affects security. Star topology is easier to secure because all traffic passes through the central hub where monitoring is possible.

Resources Needed:
- Yarn pieces
- Index cards
- Scissors
- Lab topology diagram
- Textbook
- Floor space for models

Differentiation

Approaching: Provide pre-cut cards and guided instructions for building each topology model step by step
On Level: Students create topology models and diagrams following standard textbook descriptions
Advanced: Advanced students research hybrid topologies and explain how combining topologies improves network performance
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Topology drawing test and comparison chart
CT Skill Assessed: Pattern recognition identifying relationships between network structure and performance

Dormitory — Retrieval (Paper)

Look at the lights in your dormitory corridor. Are they arranged in a line, circle, or scattered pattern? Which network topology does this arrangement resemble?

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Imagine you need to send a large textbook to your friend in another KREIS school by courier. The courier company has small boxes. How would you pack the book? What information would you write on each box to ensure it reaches the correct destination?

Students students discuss breaking the book into sections, labeling each box with sender and receiver address, and trusting the courier to deliver all boxes correctly.

Explore (10 min)

Teacher Use these Lego blocks to simulate data transmission. The tower represents a large file. Break it into smaller stacks like packets. Label each stack with a number. Now arrange them at the destination in the correct order using the numbers.

Students students experience how large data is divided into packets, transmitted separately, and reassembled at destination. They notice packets may arrive out of order.

Explain (10 min)

Teacher TCP or Transmission Control Protocol breaks data into small packets, numbers them, and ensures all packets arrive correctly at the destination. If any packet is lost TCP requests retransmission. IP or Internet Protocol handles addressing and routing. It puts the sender and receiver addresses on each packet so routers know where to send them.

Students students create a flow diagram showing data being broken into packets at source, traveling through routers, and being reassembled at destination. They label TCP and IP functions.

Elaborate (10 min)

Teacher The TCP/IP model has four layers. Application layer handles email, web browsing, and file transfer. Transport layer handles TCP and UDP. Internet layer handles IP addressing. Network access layer handles physical transmission. Match each layer to its function.

Students students create a four-layer diagram and place example activities in each layer. They discuss how layers work together like different departments in a company.

Evaluate (5 min)

Teacher Explain in your own words how TCP ensures reliable data transmission. Describe the role of IP in delivering packets to the correct destination. Draw the four-layer TCP/IP model.

Students students demonstrate understanding through clear explanations and accurate diagrams. They connect concepts to real Internet usage.

Cross-Curricular Connections

Mathematics: Calculate the number of packets needed to send a four megabyte file if each packet contains fifty kilobytes of data.
Science: Understand how electrical or optical signals carry packet data through physical cables at the speed of light.
Language: Write a step-by-step process description explaining how an email travels from one computer to another using TCP/IP.

Digital Citizen Reminder: TCP/IP protocols can be intercepted by hackers. Always use encrypted connections like HTTPS to protect your data during transmission across the Internet.

Resources Needed:
- Lego blocks
- Numbered labels
- Paper for packet simulation
- Textbook diagrams
- Whiteboard
- Colored markers

Differentiation

Approaching: Provide a simplified packet diagram with pre-labeled layers and guided matching activities
On Level: Students complete standard labeling and description activities from the textbook
Advanced: Advanced students research UDP protocol and compare its speed versus TCP reliability trade-off
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Packet tracing worksheet and TCP/IP model diagram
CT Skill Assessed: Decomposition breaking complex processes into smaller manageable steps

Dormitory — Retrieval (Paper)

Think about a message you sent on your phone today. How many steps might it have taken to reach the other person? Write at least five steps.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Think about a post office. Letters arrive from many places and need to be sorted and sent to the correct addresses. A network works similarly with special devices. Can you guess what devices help sort and direct data in a network?

Students students guess devices like routers, switches, and modems. They draw parallels between post office sorting and network device functions.

Explore (10 min)

Teacher Examine the network equipment in our computer lab. Identify the modem that connects to the Internet line, the router that distributes the connection, and the switches that connect individual computers. Trace the cable from the wall socket to a computer.

Students students physically locate and trace network devices. They observe the connection path from Internet line through modem to router to switch to computer.

Explain (10 min)

Teacher A modem converts digital signals from your computer to analog signals for telephone lines and back. A router directs data packets between different networks like from your school network to the Internet. A switch connects multiple devices within a LAN and directs data to the correct device. An access point provides wireless connectivity. A hub broadcasts data to all devices unlike a switch which sends only to the intended device.

Students students create a device diagram showing how each device connects in sequence. They write the function of each device and draw signal flow arrows.

Elaborate (10 min)

Teacher Design a complete network for a new KREIS school building with twenty computers, a printer, and Internet access. Choose and place the correct network devices. Explain why each device is needed and where it should be located.

Students students propose modem for Internet conversion, router for distribution, switches for connecting computers, and print server for shared printing. They justify placement based on cable length and signal strength.

Evaluate (5 min)

Teacher Match each device to its function. Router directs packets between networks. Switch connects devices in LAN. Modem converts signals. Access point provides Wi-Fi. Draw the device connection sequence.

Students students correctly match all devices and draw accurate connection diagrams showing the path from Internet to end device.

Cross-Curricular Connections

Mathematics: Calculate the maximum number of computers that can connect to a switch with sixteen ports and two switches with eight ports.
Science: Explain how modems convert digital signals to analog using modulation and demodulation principles.
Language: Write a user manual paragraph explaining how to connect a new computer to the school network step by step.

Digital Citizen Reminder: Network devices should be physically secured. Never unplug or modify network equipment without permission as it can disrupt connectivity for all users.

Resources Needed:
- Modem sample
- Router image
- Switch diagram
- Lab network layout
- Cable samples
- Textbook

Differentiation

Approaching: Provide device cards with pictures and simplified function descriptions for matching activities
On Level: Students complete standard device identification and function description tasks
Advanced: Advanced students research load balancers and firewalls and their roles in enterprise networks
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Device identification quiz and network design activity
CT Skill Assessed: Systems thinking understanding how components work together in a network

Dormitory — Retrieval (Paper)

Find the Wi-Fi router in your hostel block. Do not touch it. Draw a simple diagram showing where it is located and guess how many rooms it covers.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher How does music travel from your phone to your Bluetooth earphones without any wires? How does the Internet reach your laptop in the hostel without a cable plugged into it? What invisible force carries these signals?

Students students discuss radio waves, Bluetooth, and Wi-Fi. They wonder how information travels through air without physical connection.

Explore (10 min)

Teacher Use the school Wi-Fi settings to view available networks. Note the network name, signal strength, and security type. Now compare with a neighbor's phone showing different available networks. What networks appear on both devices?

Students students observe that both devices see the same school Wi-Fi network. They notice signal strength varies based on distance from the access point.

Explain (10 min)

Teacher Wireless networks use radio waves to transmit data between devices. Wi-Fi follows IEEE 802.11 standards. 802.11n offers speeds up to six hundred Mbps. 802.11ac offers speeds up to three point five Gbps. 802.11ax or Wi-Fi 6 offers even faster speeds with better performance in crowded areas. Wi-Fi range typically covers thirty to one hundred meters indoors.

Students students compare Wi-Fi standards in a table noting speed, frequency, and range. They understand why newer standards are better for streaming and gaming.

Elaborate (10 min)

Teacher Test the Wi-Fi signal strength at different locations in the school campus. Measure signal at the computer lab, library, dining hall, and hostel. Create a signal strength map showing the best and weakest coverage areas.

Students students walk around campus with devices recording signal bars. They identify dead zones and suggest where additional access points could improve coverage.

Evaluate (5 min)

Teacher List three advantages of wireless networks over wired networks. Describe two security measures used to protect Wi-Fi networks. Explain why Wi-Fi signal becomes weaker through walls.

Students students explain convenience and mobility advantages. They describe WPA encryption and password protection. They understand that physical barriers block radio waves.

Cross-Curricular Connections

Mathematics: Calculate the time to download a one hundred MB file at Wi-Fi 5 speed of one Gbps versus Wi-Fi 4 speed of one hundred fifty Mbps.
Science: Explain how radio waves at two point four GHz and five GHz frequencies differ in speed and wall penetration ability.
Language: Write a report comparing wireless and wired connections for different school activities like exams, video classes, and file sharing.

Digital Citizen Reminder: Always connect to trusted Wi-Fi networks only. Public Wi-Fi in bus stands and cafes can be hacked. Never perform banking or enter passwords on public networks.

Resources Needed:
- School Wi-Fi access
- Smartphones for signal testing
- Signal strength apps
- Coverage map template
- Textbook

Differentiation

Approaching: Provide a pre-made signal strength chart with picture icons for recording measurements
On Level: Students conduct standard Wi-Fi coverage testing and record results in a table
Advanced: Advanced students research mesh Wi-Fi systems and present how they eliminate dead zones in large buildings
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Wi-Fi coverage map and signal comparison report
CT Skill Assessed: Experimental testing gathering data through measurement and observation

Dormitory — Retrieval (Paper)

Check the Wi-Fi signal strength in your dormitory room tonight. Record the number of bars. Move to different spots in the room and note if the signal changes.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher When you visit a website you see either http or https in the address bar. What do these letters mean? Why do some websites show a green lock symbol while others show a warning? Think about the rules that govern Internet communication.

Students students notice the difference between http and https websites. They observe lock symbols on banking sites and warning signs on insecure sites.

Explore (10 min)

Teacher Open five different websites on the school computer. Check the address bar of each. Categorize them into http and https groups. Now try to access an FTP site to download a free software. Observe how the connection process differs from regular web browsing.

Students students find most educational sites use https while some simple sites still use http. They experience FTP as a different type of file access interface.

Explain (10 min)

Teacher Protocols are rules that govern communication. HTTP or Hypertext Transfer Protocol sends data in plain text without encryption. HTTPS adds SSL or TLS encryption making it secure. FTP or File Transfer Protocol is used for uploading and downloading files between servers and computers. SMTP or Simple Mail Transfer Protocol is used for sending emails. POP3 and IMAP are used for receiving emails.

Students students create a protocol chart with columns for name, full form, purpose, security level, and port number. They highlight that HTTPS is essential for secure transactions.

Elaborate (10 min)

Teacher Simulate a web request. One student acts as the browser sending an HTTP request. Another acts as the server responding with a web page. A third student acts as a hacker trying to read the message. Use plain text first then use encoded words to simulate HTTPS encryption.

Students students experience the difference between unencrypted and encrypted communication. The hacker can easily read plain text but cannot decode encrypted messages.

Evaluate (5 min)

Teacher Match each protocol to its function. HTTP for web browsing, HTTPS for secure browsing, FTP for file transfer, SMTP for sending email. Explain why online shopping requires HTTPS instead of HTTP.

> **Students** students correctly match protocols and explain that HTTPS protects credit card and personal information during online transactions.

Cross-Curricular Connections

Mathematics: Calculate the port numbers used by common protocols. HTTP uses port eighty, HTTPS uses port four forty-three, FTP uses port twenty-one, SMTP uses port twenty-five.
Science: Explain how SSL encryption works by converting readable text into scrambled data using mathematical algorithms.
Language: Write a warning notice for school students explaining why they should only visit HTTPS websites when entering passwords.

Digital Citizen Reminder: Always check for HTTPS before entering passwords or personal information. The padlock symbol in the address bar means your connection is encrypted and safe from eavesdropping.

Resources Needed:

- Computer with browser
- FTP client demo
- Protocol comparison chart
- SSL certificate example
- Textbook

Differentiation

Approaching: Provide a protocol matching worksheet with pictures and simplified descriptions
On Level: Students complete standard protocol identification and comparison activities
Advanced: Advanced students research DNS protocol and explain how website names translate to IP addresses
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Protocol identification quiz and encryption demonstration
CT Skill Assessed: Rule-based understanding applying protocols as sets of communication rules

Dormitory — Retrieval (Paper)

Check the address bar of any website you visit tonight. Write whether it is HTTP or HTTPS and explain what the difference means for your safety.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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(..) => ..
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Engage (5 min)

Teacher Your friend tells you that someone hacked into their email and read all their private messages. How do you think this happened? What would you do to prevent it from happening to you?

Students students discuss weak passwords, suspicious links, and public Wi-Fi dangers. They realize security requires both technology and personal vigilance.

Explore (10 min)

Teacher Examine these examples of phishing emails and fake websites on the board. Some look very real with school logos and official language. Others have spelling errors and suspicious links. Identify the red flags in each example that reveal it is fake.

Students students spot suspicious email addresses, urgent language demanding immediate action, misspelled URLs, and requests for passwords. They learn to verify sender identity.

Explain (10 min)

Teacher Network security threats include viruses that damage files, worms that spread across networks, phishing that steals passwords through fake sites, and denial of service attacks that crash websites. Protection includes strong passwords with letters numbers and symbols, firewalls that block unauthorized access, antivirus software that detects malware, and HTTPS encryption that protects data in transit.

Students students create a security guide listing five threats and five protection methods. They practice creating strong passwords and identifying phishing signs.

Elaborate (10 min)

Teacher Conduct a security audit of the school computer lab. Check if antivirus is updated, firewall is enabled, passwords are strong, and browsers are current. Write a report with recommendations for improving security.

Students students inspect lab computers and find that some have weak passwords like one two three four five six. They recommend password policies and regular updates.

Evaluate (5 min)

Teacher Classify each scenario as safe or risky. Using HTTPS for banking. Clicking links in unknown emails. Updating antivirus regularly. Sharing passwords with friends. Downloading software from official sites only.

Students students correctly identify risky behaviors. They explain why sharing passwords and clicking unknown links are dangerous practices.

Cross-Curricular Connections

Mathematics: Calculate how many possible combinations exist for an eight-character password using letters, numbers, and symbols versus a four-digit PIN.
Science: Understand how antivirus software uses signature databases to identify known malware patterns in files.

Language: Write a school newsletter article teaching younger students about password safety and phishing prevention.

Digital Citizen Reminder: Never share your passwords. Use different passwords for different accounts. Change your passwords every three months. Enable two-factor authentication wherever possible.

Resources Needed:
- Phishing email samples
- Password strength checker
- Antivirus software demo
- Security checklist
- Textbook

Differentiation

Approaching: Provide a picture-based security matching activity with threat icons and solution icons
On Level: Students complete standard threat identification and protection method matching
Advanced: Advanced students research encryption algorithms and present how AES protects data
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Security awareness quiz and password creation activity
CT Skill Assessed: Risk assessment identifying vulnerabilities and proposing solutions

Dormitory — Retrieval (Paper)

Check if your phone has a screen lock. If yes, describe the type. If no, set one up tonight using a pattern or six-digit PIN.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

> **Teacher** You have five hundred photos on your phone but the storage is full. You could delete some photos or move them somewhere that never gets full and you can access from any device. Where would you move them?

Students students think of external hard drives and cloud storage. They realize cloud storage can be accessed from anywhere with Internet.

Explore (10 min)

Teacher Log into the school Google account and explore Google Drive. Notice how much free storage is available. Create a folder, upload a text file, and share it with a classmate. Observe how the file appears on your classmate's device immediately.

Students students experience instant file synchronization across devices. They see that cloud storage requires Internet but provides accessibility and sharing features.

Explain (10 min)

Teacher Cloud computing means storing and accessing data and programs over the Internet instead of your computer's hard drive. Cloud storage services include Google Drive with fifteen GB free, Microsoft OneDrive with five GB free, and Dropbox with two GB free. Advantages include access from anywhere, automatic backup, easy sharing, and no hardware to carry. Disadvantages include need for Internet, privacy concerns, and storage limits.

Students students create a comparison table of cloud storage services with columns for free space, features, and best use cases. They understand trade-offs between convenience and control.

Elaborate (10 min)

Teacher Imagine KREIS school wants to move all student records to the cloud. What are the benefits and risks? Design a plan for safe cloud migration including what data to move, what security measures to implement, and how to train staff.

Students students propose benefits like teacher access from anywhere and automatic backup. They identify risks like data breaches and propose encryption and access controls.

Evaluate (5 min)

Teacher Answer these questions. What is cloud storage? Name two cloud services. List two advantages over USB drives. Explain when local storage is better than cloud storage.

Students students define cloud storage correctly, name services, list accessibility advantages, and explain that large files or offline work need local storage.

Cross-Curricular Connections

Mathematics: Calculate the total free cloud storage available if you sign up for Google Drive, OneDrive, and Dropbox combined.
Science: Explain how data centers use cooling systems and backup power to keep cloud servers running twenty-four hours every day.
Language: Write a comparison essay about traditional filing cabinets versus digital cloud storage for a school records system.

Digital Citizen Reminder: Cloud storage providers can access your files. Read their privacy policies. Never upload sensitive personal documents like Aadhaar details to free cloud services without proper security.

Resources Needed:
- Google account access
- Cloud storage comparison chart
- USB drive sample
- Textbook
- Projector for demo

Differentiation

Approaching: Provide step-by-step picture guides for uploading files to Google Drive with screenshots
On Level: Students complete standard cloud storage comparison and file upload activities
Advanced: Advanced students research cloud computing models like SaaS, PaaS, and IaaS with real examples
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Cloud storage comparison worksheet and file management task
CT Skill Assessed: Comparative evaluation weighing benefits against limitations of different storage methods

Dormitory — Retrieval (Paper)

Check how much storage is left on your phone. If it is almost full, identify five photos or apps you could move to cloud storage to free up space.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Our school wants to build a new computer lab with thirty computers, a printer, Internet access, and Wi-Fi for the whole campus. You have been hired as the network consultant. What will you need to plan and design?

Students students brainstorm requirements including devices, cables, topologies, security, and budget. They realize networking projects need careful planning.

Explore (10 min)

Teacher Examine the network designs created by previous batches. Compare three different designs for a ten-computer lab. One uses only wireless, one uses only wired, and one uses a hybrid approach. Evaluate which design is best and why.

Students students identify that hybrid design offers best balance of speed and flexibility. They note wireless is convenient but wired is more reliable for heavy use.

Explain (10 min)

Teacher A network project requires planning several components. Hardware includes computers, servers, switches, routers, and cables. Software includes operating systems and network management tools. Security includes firewalls, antivirus, and access controls. Documentation includes network diagrams, IP address allocation, and maintenance schedules. Budget includes one-time setup cost and annual maintenance cost.

Students students create a project checklist with all components. They understand that each component must be planned before implementation begins.

Elaborate (10 min)

Teacher Work in teams to design a complete network for a new KREIS school wing with a library, computer lab, and staff room. Create a network diagram, list all required equipment with quantities, estimate costs, and write a security policy document.

Students teams produce complete network plans with diagrams, equipment lists, and budgets. They present their designs to the class for peer review.

Evaluate (5 min)

Teacher Present your network design to the class. Explain your topology choice, device selection, and security measures. Answer questions from classmates and the teacher about your design decisions.

Students students present confidently and defend their choices with reasoning. They receive feedback on completeness, cost-effectiveness, and security of their designs.

Cross-Curricular Connections

Mathematics: Calculate the total budget for networking equipment including computers at thirty-five thousand each, switches at five thousand each, and cables at two thousand per room.
Science: Research how solar power can provide electricity for network equipment in rural KREIS schools without reliable power supply.
Language: Write a formal project proposal document with introduction, requirements, design, budget, and conclusion sections.

Digital Citizen Reminder: Network projects must include security planning from the beginning. Never build a network without considering how to protect student data and prevent unauthorized access.

Resources Needed:
- Network design templates
- Equipment price list
- Projector for presentation
- Chart paper for diagrams
- Textbook reference

Differentiation

Approaching: Provide partially completed network templates where students fill in equipment details and quantities
On Level: Students create complete network designs following standard planning guidelines
Advanced: Advanced students design multi-building campus networks with redundant connections and failover systems
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Group presentation and network design portfolio evaluation
CT Skill Assessed: Project management planning and executing a complex technical project

Dormitory — Retrieval (Paper)

Draw a simple map of your hostel floor. Mark where Wi-Fi access points should be placed for best coverage. Think about which walls signals must pass through.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Chapter 2 — Advanced Graphics

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Engage (5 min)

Teacher Look at this poster on the wall and this photo on the screen. One is made of tiny squares called pixels and the other is made of mathematical lines. What happens when you zoom into each one?

Students students observe that zooming into the pixel image makes it blurry while the line image stays sharp. They realize images are created in different ways.

Explore (10 min)

Teacher Open these three image files on the computer. A JPEG photo, a PNG logo, and a SVG icon. Zoom into each one to two hundred percent. Observe what happens to the quality. Note the file sizes of each format.

Students students discover that JPEG loses quality when zoomed while PNG stays crisp for sharp edges. SVG remains perfect at any zoom level. They notice PNG files are larger.

Explain (10 min)

Teacher Raster images like JPEG and PNG are made of a grid of pixels. Each pixel has a color. When you enlarge a raster image the pixels become visible and quality decreases. Vector images like SVG are made of mathematical paths and points. They can be scaled to any size without losing quality. JPEG is best for photos. PNG is best for graphics with transparency. SVG is best for icons and logos.

Students students create a comparison chart of image formats with columns for type, best use, transparency support, and file size. They understand when to use each format.

Elaborate (10 min)

Teacher Think about all the images you see daily. School posters, textbook illustrations, phone photos, website icons, and social media posts. Classify each as raster or vector and explain which format would be best for creating it.

Students students categorize images and realize that photos are raster while logos and icons should be vector. They plan to use appropriate formats for their graphics projects.

Evaluate (5 min)

Teacher Answer these questions. What is the difference between raster and vector images? Name two raster formats and two vector formats. When would you choose PNG over JPEG?

Students students explain pixel versus mathematical composition, name formats correctly, and describe PNG transparency advantage for logos.

Cross-Curricular Connections

Mathematics: Calculate the total pixels in a photo that is four thousand by three thousand pixels. Express the result in megapixels.
Science: Explain how digital cameras capture light using sensors that convert photons into electrical signals stored as pixel values.
Language: Write a guide explaining to younger students when to save their drawings as JPEG versus PNG based on the type of image.

Digital Citizen Reminder: Respect copyright when using images. Always check if images are free to use or require permission. Give credit to original creators when using their work.

Resources Needed:
- Image samples in different formats
- Computer with GIMP installed
- Projector
- Comparison chart template
- Textbook

Differentiation

Approaching: Provide pre-made image samples with labels and a simplified format comparison sheet
On Level: Students complete standard format identification and comparison activities

Advanced: Advanced students research image compression algorithms and explain lossy versus lossy compression mathematically
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Image format identification quiz and comparison chart
CT Skill Assessed: Classification understanding different data types and their appropriate applications

Dormitory — Retrieval (Paper)

Open the camera app on your phone and take two photos. Check the file details. Write the file format, dimensions, and file size of each photo.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Have you ever used a paint program on a computer? How is it different from painting on paper? What tools would you expect to find in a digital painting program?

Students students mention undo button, color picker, brushes, and eraser. They realize digital painting offers tools impossible with physical paint.

Explore (10 min)

Teacher Launch GIMP on the school computer. Explore the interface without any instructions. Find the menu bar at the top, the toolbox on the left, the layers panel on the right, and the image canvas in the center. Try clicking different tools and see what happens.

Students students discover GIMP has multiple panels and tools. They find that clicking tools changes the cursor and options bar shows tool settings.

Explain (10 min)

Teacher GIMP or GNU Image Manipulation Program is free image editing software. The interface has the image window in the center, the toolbox on the left with selection tools, paint tools, and transform tools, the menu bar on top with File Edit View and Image menus, and dockable dialogs on the right for layers, channels, and paths. You can rearrange panels by dragging them.

Students students label a GIMP interface diagram with all major components. They practice opening and closing panels and rearranging the workspace.

Elaborate (10 min)

Teacher Set up your GIMP workspace for the graphics projects we will do this chapter. Create a custom workspace with the toolbox, layers panel, and tool options visible. Save this layout using Windows Dockable Dialogs.

Students students customize their workspace and learn to reset it if confused. They appreciate having a consistent workspace for efficient editing.

Evaluate (5 min)

Teacher Identify each GIMP interface element. Create a new image with width eight hundred pixels, height six hundred pixels, and resolution three hundred DPI. Save it as a XCF file.

Students students correctly locate all interface elements and create images with specified settings. They understand XCF is GIMP's native format.

Cross-Curricular Connections

Mathematics: Calculate the file size of a blank image that is eight hundred by six hundred pixels with three color channels at eight bits per channel.
Science: Explain how computer monitors display images using red green and blue sub-pixels that combine to create millions of colors.
Language: Write step-by-step instructions for a younger student on how to open GIMP and create their first blank canvas.

Digital Citizen Reminder: GIMP is free and open-source software. Respect software licenses and do not redistribute copyrighted materials created with GIMP without permission.

Resources Needed:
- Computers with GIMP installed
- GIMP interface guide
- Projector for demonstration
- Practice image files
- Textbook

Differentiation

Approaching: Provide a GIMP interface map with numbered parts for students to label and match
On Level: Students complete standard interface exploration and image creation activities
Advanced: Advanced students explore GIMP scripting and create simple automated workflows using Script-Fu
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Interface navigation checklist and image creation task
CT Skill Assessed: Tool proficiency navigating complex software interfaces efficiently

Dormitory — Retrieval (Paper)

Practice opening GIMP on any available computer tonight. Try to find the color picker tool and change the foreground color to red.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Imagine you want to cut out a circle from a colored paper to paste onto another paper. How would you do it with scissors? Now think about doing the same thing digitally on a photograph. What tool would you need?

Students students think of scissors and templates for physical cutting. They realize digital selection tools act like virtual scissors for precise cutting.

Explore (10 min)

Teacher Open a photo in GIMP. Use the rectangular selection tool to select a portion of the image. Then use the elliptical tool to select a circular area. Now try the free select tool to trace around an irregular shape. Notice how each tool creates different selection boundaries.

Students students experience different selection shapes and difficulty levels. They find rectangular is easiest while free select requires careful mouse control.

Explain (10 min)

Teacher Selection tools define which part of an image you want to edit. Rectangle select creates square or rectangular selections. Ellipse select creates circular or oval selections. Free select lets you draw any shape by clicking points. The fuzzy select or magic wand selects areas of similar color. Selection modifiers let you add to, subtract from, or intersect with existing selections using the tool options.

Students students practice each selection tool on sample images. They create complex selections by combining rectangular and elliptical shapes using add and subtract modes.

Elaborate (10 min)

Teacher Using only selection tools, create a selection that isolates just the face of a person in a group photograph. You may need to combine multiple selection tools and use zoom for precision. Practice patience and use Ctrl plus Z to undo mistakes.

Students students develop precision by combining rectangle select for initial area, ellipse select for head shape, and free select for hair details. They learn zooming helps accuracy.

Evaluate (5 min)

Teacher Complete these tasks. Select a rectangular area and fill it with red color. Select an elliptical area and delete it to create a hole. Combine rectangular and elliptical selections to create a crescent moon shape.

Students students demonstrate tool mastery by creating accurate selections and combining them creatively. Moon shape requires subtracting circle from rectangle.

Cross-Curricular Connections

Mathematics: Calculate the area of a rectangular selection that is two hundred by one hundred fifty pixels. What percentage of a eight hundred by six hundred image does it cover?
Science: Explain how the magic wand tool uses color similarity algorithms to determine which adjacent pixels to include in a selection.
Language: Write a tutorial explaining the difference between each selection tool with examples of when to use each one.

Digital Citizen Reminder: Selection skills help you properly credit and crop images when creating projects. Always ensure you have permission to modify and use images.

Resources Needed:
- Sample photographs for practice
- GIMP installed on computers
- Selection tool guide handout
- Projector
- Textbook

Differentiation

Approaching: Provide simplified images with clear boundaries for easier selection practice
On Level: Students complete standard selection exercises using all tool types
Advanced: Advanced students use the scissors select tool and foreground select tool for complex object extraction
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Selection accuracy test with practice images
CT Skill Assessed: Precision control using fine motor skills for accurate digital manipulation

Dormitory — Retrieval (Paper)

Look at objects in your room. Pick one irregularly shaped object and think about how you would trace around it using the free select tool in GIMP.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Think about a glass painting where you paint the background on one glass, middle ground on another glass, and foreground on a third glass. When you stack them you see the complete picture. How is this like layers in image editing?

Students students realize that layers stack like glass sheets allowing independent editing of each element without affecting others.

Explore (10 min)

Teacher Open a GIMP project with multiple layers. Toggle the eye icon next to each layer to hide and show it. Notice how each layer contains different elements. Change the opacity of one layer and see the layers below become visible through it.

Students students experience how layers work independently. They find that reducing opacity creates transparency effects and hiding layers reveals what is underneath.

Explain (10 min)

Teacher Layers are like transparent sheets stacked on top of each other. Each layer can contain different image elements. You can edit one layer without affecting others. Layer opacity controls transparency from fully opaque to fully invisible. Blending modes like multiply, screen, and overlay change how layers interact visually. Layer masks hide or reveal parts of a layer without deleting pixels. Layer groups organize related layers into folders.

Students students practice creating layers, adjusting opacity, and trying different blending modes. They create a simple composition using at least four layers with proper names.

Elaborate (10 min)

Teacher Create a composite image with a background landscape, a middle ground with trees, and a foreground with a person. Use separate layers for each element. Apply different blending modes to see creative effects. Use a layer mask on the person layer to blend them naturally into the scene.

Students students build multi-layer compositions understanding how each element can be adjusted independently. They discover creative blending mode effects.

Evaluate (5 min)

Teacher Describe what layers are and why they are useful. Explain the difference between opacity and blending modes. Create a project with at least five named layers demonstrating different features.

Students students articulate layer benefits clearly and demonstrate practical skills in creating organized multi-layer projects.

Cross-Curricular Connections

Mathematics: Calculate the number of possible layer combinations when stacking four layers each with three blending mode options.
Science: Explain how digital compositing uses alpha channels to determine pixel transparency values from zero to two hundred fifty-five.
Language: Write a creative story describing a scene and then plan which elements would go on separate layers if you were to illustrate it.

Digital Citizen Reminder: Layers help you organize your work professionally. Proper layer naming and organization makes collaboration easier when sharing project files with classmates.

Resources Needed:
- GIMP with layer examples
- Sample multi-layer projects
- Layer blending mode reference
- Textbook
- Computer access

Differentiation

Approaching: Provide a layer building exercise with pre-made elements that students stack and adjust
On Level: Students create standard multi-layer compositions following step-by-step instructions
Advanced: Advanced students explore layer effects like drop shadows and gradients for professional-looking results
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Layer composition project with named layers
CT Skill Assessed: Hierarchical organization managing multiple components in a structured system

Dormitory — Retrieval (Paper)

Think about your dormitory room. If you were to draw it in layers, what would go on the background layer, middle layer, and foreground layer?

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Look at this blurry photograph. Can you make it sharp again? Now look at this sharp photo. Can you make it look like a painting? What magical tools can transform images in these ways?

Students students wonder if blur can be reversed and how photos become paintings. They discover filters are powerful transformation tools.

Explore (10 min)

Teacher Open a photograph in GIMP. Go to Filters menu and try Gaussian Blur with radius five. Then try Unsharp Mask to sharpen. Try the Artistic Oilify filter. Try the Distorts iWarp filter. Observe how each filter changes the image differently.

Students students experience that blur softens details, sharpening enhances edges, oilify creates a painting effect, and iWarp creates 变形 effects. They note some filters have adjustable settings.

Explain (10 min)

Teacher Filters apply mathematical transformations to pixel values. Blur filters like Gaussian Blur reduce detail and create soft effects useful for backgrounds. Sharpen filters like Unsharp Mask enhance edges and details useful for improving focus. Artistic filters like Oilify and Charcoal create hand-drawn effects. Distort filters like iWarp and Waves bend and twist image elements. Each filter has settings you can adjust for different intensity levels.

Students students create a filter reference guide listing each filter type, its effect, common settings, and when to use it. They understand filters are tools not toys.

Elaborate (10 min)

Teacher Take a portrait photograph and apply a sequence of filters to create an oil painting effect. First apply slight blur to reduce noise. Then apply Oilify filter. Then adjust colors using Color Balance. Save each step as a separate layer to show your filter progression.

Students students learn that combining filters creates more convincing effects than using one filter alone. They practice building effects progressively.

Evaluate (5 min)

Teacher Apply Gaussian Blur to a background layer while keeping the subject sharp. Create a pencil sketch effect using Desaturate and Edge Detect filters. Explain why you chose each filter for its specific purpose.

Students students demonstrate practical filter knowledge by selecting appropriate filters for specific creative goals and explaining their reasoning.

Cross-Curricular Connections

Mathematics: Calculate the mathematical difference between a pixel value of one hundred fifty before and after applying a sharpen filter that increases contrast by twenty percent.
Science: Explain how Gaussian Blur uses a bell curve mathematical function to average pixel values with their neighbors.
Language: Write an art critique comparing a original photograph with its filtered version analyzing mood and visual impact changes.

Digital Citizen Reminder: Filters can be used to misleadingly alter photos. Always disclose when images have been significantly filtered, especially in news or educational contexts.

Resources Needed:
- Sample photographs
- GIMP filter menu reference
- Filter effect comparison chart
- Projector
- Textbook

Differentiation

Approaching: Provide a filter matching activity with before and after images for students to identify which filter was applied
On Level: Students complete standard filter application exercises with specific settings
Advanced: Advanced students explore Script-Fu to create custom filter chains for batch processing multiple images
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Filter application portfolio with before and after comparisons
CT Skill Assessed: Creative application using technical tools to achieve artistic vision

Dormitory — Retrieval (Paper)

Find a photo on your phone. Think about what filter would make it look like a painting. Try applying a similar effect using any photo app on your phone.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

> **Teacher** Look at this photo taken in the evening. It looks too dark and yellow. This other photo was taken in shade and looks too blue. How can we fix these color problems to make the photos look natural?

> **Students** students notice that lighting conditions create color casts. They wonder how to restore natural colors in poorly lit photographs.

Explore (10 min)

> **Teacher** Open a dark photograph in GIMP. Use Colors Brightness-Contrast to increase brightness. Then open a photo with a yellow color cast and use Colors Color Balance to add blue. Try Colors Hue-Saturation to make colors more vibrant. Compare before and after results.

> **Students** students experience that brightness reveals hidden details, color balance corrects temperature, and saturation makes colors pop. They learn each tool serves a specific purpose.

Explain (10 min)

> **Teacher** Color correction fixes problems in photographs. Brightness increases or decreases overall lightness. Contrast increases the difference between light and dark areas. Saturation controls color intensity. Color Balance adjusts the mix of red, green, and blue to correct color casts. Curves gives precise control over tonal range by adjusting specific brightness levels. Levels shows a histogram of tonal distribution and lets you set black and white points.

> **Students** students practice each adjustment tool on sample images. They create a before and after comparison showing improvement in exposure and color accuracy.

Elaborate (10 min)

> **Teacher** Take three poorly exposed photographs intentionally. One too dark, one too bright, and one with a strong color cast. Use different color correction tools to fix each one. Document your adjustment steps and settings for each correction.

> **Students** students develop judgment about which corrections to apply and in what order. They learn that sometimes multiple adjustments are needed for the best result.

Evaluate (5 min)

Teacher Correct this overexposed photograph using appropriate tools. Adjust the color balance of this indoor photo to remove the orange cast. Increase the vibrance of this dull landscape without making skin tones unnatural.

Students students demonstrate understanding of color theory by making corrections that look natural and enhance the image without over-processing.

Cross-Curricular Connections

Mathematics: Calculate the percentage increase in brightness when a pixel value changes from one hundred to one hundred forty.
Science: Explain how the human eye perceives color differently under warm tungsten light versus cool fluorescent light.
Language: Write a photo editing guide for beginners explaining when to use brightness versus contrast versus color balance.

Digital Citizen Reminder: Color correction should enhance truth not distort it. Be honest when editing photos for school projects and always mention adjustments made.

Resources Needed:
- Underexposed photos
- Overexposed photos
- Color cast examples
- GIMP color tools
- Histogram reference
- Textbook

Differentiation

Approaching: Provide before and after comparison images with sliders showing approximate adjustment values
On Level: Students complete standard color correction exercises on provided practice images
Advanced: Advanced students use the Curves tool with the histogram to make precise point-by-point tonal adjustments
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Color correction exercise with before and after comparisons
CT Skill Assessed: Analytical adjustment modifying parameters based on visual feedback

Dormitory — Retrieval (Paper)

Take a photo in your dormitory tonight with the lights on. Does it look too yellow or too blue? Write what color correction you would apply to fix it.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Look at these two posters. One uses small plain text and the other uses large colorful creative text. Which one catches your eye first? What makes the second poster more attractive?

Students students notice that creative text design makes posters more appealing. They identify font choice, size, color, and placement as key design elements.

Explore (10 min)

Teacher Open GIMP and select the text tool. Click on the canvas and type your name. Try changing the font, size, and color in the tool options. Now create a second text layer with different settings. Try placing text over a photograph and see how it looks.

Students students experiment with text properties and discover that font choice dramatically changes the message feel. They find that text over images needs careful color choice for readability.

Explain (10 min)

Teacher Typography is the art of arranging text for readability and visual appeal. Font choice affects mood. Serif fonts like Times feel formal. Sans-serif fonts like Arial feel modern. Script fonts feel decorative. Font size determines hierarchy. Headlines should be larger than body text. Text color should contrast with background for readability. Letter spacing or kerning affects how text feels. Line spacing or leading affects how comfortable text is to read.

Students students analyze typography in school posters and textbooks. They create a typography guide listing font choices for different purposes like formal, creative, and fun.

Elaborate (10 min)

Teacher Design a title poster for your class event using GIMP. Choose appropriate fonts for the title and subtitle. Use at least three different text effects like drop shadow, outline, or gradient fill. Ensure the text is readable against the background.

Students students apply typography principles to create effective poster designs. They learn that text effects should enhance readability not reduce it.

Evaluate (5 min)

Teacher Create a text composition with at least three different fonts that work together harmoniously. Apply a text effect to make the title stand out. Explain your font choices and why they suit the message.

Students students demonstrate understanding of font pairing and hierarchy. They justify their choices based on mood and readability principles.

Cross-Curricular Connections

Mathematics: Calculate the ideal line length for body text. Research suggests forty-five to seventy-five characters per line for comfortable reading. How many characters fit in a three hundred pixel wide text box at twelve point font?
Science: Explain how digital fonts are created using vector outlines called glyphs that define each character shape at any size.
Language: Write a comparison of traditional handwritten calligraphy and digital typography analyzing the advantages and limitations of each.

Digital Citizen Reminder: Respect font licenses. Many fonts are free for personal use but require licenses for commercial use. Use Google Fonts for free fonts you can safely use in projects.

Resources Needed:
- GIMP text tool
- Font samples
- Typography poster examples
- Google Fonts website
- Textbook

Differentiation

Approaching: Provide font pairing suggestions and pre-made text effect templates for practice
On Level: Students create standard typography exercises following design principles
Advanced: Advanced students explore text on path effects and perspective text transformations
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Typography design project with written justification
CT Skill Assessed: Design thinking balancing aesthetics with functional communication

Dormitory — Retrieval (Paper)

Look at the text on your textbook cover. How many different fonts are used? What mood does each font choice create?

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Look at this magazine cover. Do you think the model really looks like that in real life? What tools might have been used to make the skin look so perfect? Is it ethical to change how people look in photos?

Students students discuss photo retouching in media and debate ethics. They wonder where the line is between enhancement and deception.

Explore (10 min)

Teacher Open a portrait photo in GIMP. Find a small blemish on the face. Use the heal tool to click on the blemish and remove it. Now find an unwanted object in the background. Use the clone stamp tool to paint over it with a nearby texture. Compare before and after.

Students students experience that heal tool blends texture automatically while clone stamp copies exactly. They practice on small areas first for confidence.

Explain (10 min)

Teacher Photo retouching improves images by removing imperfections. The heal tool samples texture from surrounding areas and blends it to cover blemishes. The clone stamp copies pixels from one area to another precisely. The blur tool can soften harsh lines. Color adjustment can improve skin tone. Important principles include keeping skin texture natural, not over-smoothing faces, and preserving the person's actual appearance.

Students students practice retouching on sample portraits. They learn that good retouching is invisible while bad retouching looks plastic and fake.

Elaborate (10 min)

Teacher Retouch a school group photograph. Remove any distracting background elements using clone stamp. Clean up minor blemishes on faces using heal tool. Adjust brightness so all faces are clearly visible. The goal is to improve the photo while making it look unedited.

Students students practice subtle retouching that enhances without obvious manipulation. They understand that the best retouching is unnoticeable.

Evaluate (5 min)

Teacher Retouch this portrait to remove three blemishes while keeping the natural skin texture. Remove this unwanted person from the background. Explain the ethical considerations of retouching people in photographs.

Students students demonstrate technical skill and ethical awareness. They can retouch effectively while discussing when retouching crosses ethical boundaries.

Cross-Curricular Connections

Mathematics: Calculate how many pixels need to be sampled by the clone stamp tool to cover a one hundred by fifty pixel area at one sample per pixel.
Science: Explain how the heal tool uses interpolation algorithms to blend sampled texture with surrounding color and brightness values.
Language: Write an editorial arguing for or against photo retouching in school yearbooks and educational materials.

Digital Citizen Reminder: Retouching should never be used to misrepresent people or events. In educational contexts, disclose when photos have been retouched. Do not retouch photos of others without their consent.

Resources Needed:
- Portrait photos for practice
- Retouching examples
- Ethical guidelines handout
- GIMP tools reference
- Textbook

Differentiation

Approaching: Provide photos with obvious blemishes that are easy to identify and remove
On Level: Students complete standard retouching exercises on provided practice images
Advanced: Advanced students explore frequency separation technique for professional portrait retouching
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Retouching portfolio with before and after comparisons
CT Skill Assessed: Ethical application balancing technical skill with responsible use

Dormitory — Retrieval (Paper)

Think about photos you have seen on social media. Can you identify any that might have been retouched? What signs gave it away?

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

> **Teacher** Have you seen movie posters where actors stand in impossible locations like on top of mountains or in space? How do you think these images are created? What if you could put yourself in any location in the world?

> **Students** students realize that movie posters combine multiple photographs. They get excited about creating their own composite images.

Explore (10 min)

Teacher Open two photographs in GIMP. Select an object from the first image and copy it. Paste it into the second image. Notice how the lighting and colors do not match. Try adjusting the pasted layer to blend better with the background.

Students students discover that combining images requires color and lighting matching. They find that simply pasting looks unnatural without adjustment.

Explain (10 min)

Teacher Compositing combines elements from different images into one realistic scene. Steps include selecting and extracting the subject, placing it on a new background, matching color temperature using Color Balance, adjusting brightness and contrast to match the scene, adding shadows to ground the subject, and using layer masks for soft edges. The key to believable compositing is matching light direction, color temperature, and perspective.

Students students practice compositing by placing a person from one photo onto a different background. They follow a step-by-step workflow ensuring realistic results.

Elaborate (10 min)

Teacher Create a fantasy composite image placing yourself in a famous location. Extract yourself from a school photo, place yourself on a mountain top or beach, match the lighting and colors, add a realistic shadow, and use a layer mask to blend your feet naturally into the ground.

Students students create imaginative composites while applying technical skills. They learn that patience and attention to detail create convincing results.

Evaluate (5 min)

Teacher Evaluate this composite image for realism. Identify three problems with the lighting or color matching. Create your own composite and have a classmate evaluate it for natural appearance.

Students students develop critical evaluation skills for compositing quality. They can identify mismatched elements and improve their own work through peer feedback.

Cross-Curricular Connections

Mathematics: Calculate the angle of shadows in two source images to ensure they match. If one shadow falls at thirty degrees and another at forty-five degrees, how would you adjust them?
Science: Explain how light direction affects color temperature. Direct sunlight creates warm tones while shade creates cool blue tones.
Language: Write a behind-the-scenes description of how a movie poster composite was created listing all the technical steps involved.

Digital Citizen Reminder: Composites should be labeled as manipulated images when used in educational or news contexts. Never create composites that misrepresent real events or people.

Resources Needed:
- Multiple source photographs
- Green screen sample images
- Compositing tutorial
- Color matching guide
- Textbook

Differentiation

Approaching: Provide pre-extracted subjects with matching backgrounds for simpler compositing practice
On Level: Students complete standard compositing exercises with provided source images
Advanced: Advanced students create complex multi-element composites with atmospheric effects like fog and light rays
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Composite image evaluation using realism checklist
CT Skill Assessed: Integration combining multiple data sources into coherent unified output

Dormitory — Retrieval (Paper)

Look at the view from your dormitory window. If you could place yourself anywhere in the world, where would you go? Describe what your composite photo would look like.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

> **Teacher** You see product photos on shopping websites where the item appears on a pure white background. Do you think the photo was taken with a white background or was the original background removed? How would you do this for your own photos?

> **Students** students realize that many product photos have backgrounds removed and replaced. They want to learn how to isolate objects from their original settings.

Explore (10 min)

Teacher Open a photo of a person against a busy background in GIMP. Try using the fuzzy select tool to select the background. Notice how it selects similar colored areas. Now try the foreground select tool to paint over the person and let GIMP separate them from the background.

Students students discover that simple backgrounds are easier to remove than complex ones. They find that foreground select tool gives better results for complex shapes.

Explain (10 min)

Teacher Background removal isolates the subject from its surroundings. Methods include magic wand for simple uniform backgrounds, foreground select tool for complex subjects, path tool for precise edges, and layer masks for non-destructive editing. After extraction, clean up edges using the eraser with soft brush, refine edges using Select Border, and add the subject to a new background. Save as PNG to preserve transparency.

Students students practice multiple extraction methods and learn which works best for different situations. They understand that no single method works perfectly for all images.

Elaborate (10 min)

Teacher Remove the background from a school event photograph. Place the extracted subjects onto a plain colored background for a yearbook style photo. Ensure edges are clean and no background remnants remain. Save as PNG for transparency and as JPEG with white background for printing.

Students students complete a practical background removal project. They learn to output files in appropriate formats for different uses.

Evaluate (5 min)

Teacher Compare three background removal methods on the same image. Rate each for accuracy, speed, and ease of use. Which method gave the best result for hair details?

Students students critically evaluate different approaches. They find that foreground select tool works best for people while magic wand works for simple objects.

Cross-Curricular Connections

Mathematics: Calculate the percentage of background pixels removed when extracting a subject that occupies twenty percent of the image area.
Science: Explain how image segmentation algorithms use edge detection and color clustering to automatically separate foreground from background.
Language: Write a product photography guide explaining how good lighting and simple backgrounds make background removal easier.

Digital Citizen Reminder: Only remove backgrounds from photos you own or have permission to edit. Do not extract people from photos and place them in inappropriate contexts.

Resources Needed:
- Photos with various backgrounds
- Background removal examples
- New background images
- GIMP tools
- Textbook

Differentiation

Approaching: Provide photos with solid color backgrounds for easier initial extraction practice
On Level: Students complete standard background removal exercises with increasing complexity
Advanced: Advanced students explore channel-based masking for extracting hair and transparent objects
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Background removal quality assessment with edge evaluation
CT Skill Assessed: Precision extraction isolating specific elements from complex datasets

Dormitory — Retrieval (Paper)

Find a product photo online that has a pure white background. Think about how the background was removed. Write three clues that suggest the background was edited.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

> **Teacher** Walk around the school and look at all the posters on the walls. Which poster catches your attention first? What makes it stand out from the others? Think about colors, text size, and images.

> **Students** students identify that bold colors, large text, and clear images make posters effective. They notice that cluttered posters are harder to read from a distance.

Explore (10 min)

> **Teacher** Examine three school posters on the board. Poster one has too much text in small font. Poster two has a great image but unclear message. Poster three has balanced text, image, and white space. Define what works and what does not in each poster.

> **Students** students discover that effective posters have clear hierarchy, readable text, and purposeful white space. They learn that less text with bigger font is more effective.

Explain (10 min)

Teacher Poster design follows key principles. Hierarchy guides the viewer eye from most to least important elements using size and placement. Contrast makes elements stand out using complementary colors or light versus dark. Alignment creates order by lining up elements on invisible grid lines. Repetition creates consistency by using same fonts and colors throughout. White space gives elements room to breathe and prevents visual clutter.

Students students analyze poster examples against these principles. They create a design checklist and plan their own poster before opening GIMP.

Elaborate (10 min)

Teacher Design a poster for the annual KREIS school sports day. Include event name, date, time, venue, and a relevant image. Use hierarchy to make the event name largest, add activities list in medium font, and details in smaller font. Choose colors that are energetic and sporty.

Students students create complete event posters applying all design principles. They practice planning before designing and iterating based on feedback.

Evaluate (5 min)

Teacher Evaluate your poster using the design principles checklist. Is the hierarchy clear? Is there sufficient contrast? Are elements aligned? Is there enough white space? Have a classmate review and suggest improvements.

Students students critically evaluate their own and peer work. They understand that design is an iterative process of creation and refinement.

Cross-Curricular Connections

Mathematics: Calculate the appropriate font size for a poster viewed from ten meters. Research suggests one point per thirty centimeters of viewing distance.
Science: Explain how color psychology affects poster mood. Red creates urgency, blue creates trust, green creates calm, and yellow creates optimism.
Language: Write a persuasive paragraph for your poster that motivates students to attend the sports day using compelling language.

Digital Citizen Reminder: Poster design should be inclusive and represent all students fairly. Avoid stereotypes in images and language. Ensure text is readable for students with visual impairments.

Resources Needed:
- Poster examples
- Design principles handout
- GIMP with templates
- Color palette reference
- Past sports day photos

Differentiation

Approaching: Provide pre-made poster templates where students modify text and images for their specific event
On Level: Students create posters from scratch following standard design principles
Advanced: Advanced students design poster series with consistent branding across multiple promotional materials
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Poster evaluation rubric with design principle criteria
CT Skill Assessed: Visual communication designing effective information delivery through images and text

Dormitory — Retrieval (Paper)

Look at posters in your hostel corridor tonight. Which one is most effective? Write three reasons why it works better than the others.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Throughout this chapter you have learned selection tools, layers, filters, color correction, text, retouching, compositing, background removal, and poster design. Now you will create a portfolio showcasing your best work in each area.

Students students 回顾 their progress and feel proud of accumulated skills. They realize a portfolio demonstrates their complete learning journey.

Explore (10 min)

Teacher Look at these example portfolios from professional graphic designers. Notice how each portfolio has a consistent style, clear organization, and descriptions explaining the work. What makes a portfolio impressive versus just a collection of files?

Students students identify that good portfolios tell a story of growth, show variety of skills, and include thoughtful descriptions of each piece.

Explain (10 min)

Teacher A portfolio is a curated collection of your best work that demonstrates your skills and growth. Include one example from each lesson topic with a title, description of techniques used, and what you learned. Organize work in logical order from simple to complex. Create a cover page with your name and class. Add an introduction explaining your learning journey. Export as a PDF for easy sharing.

Students students plan their portfolio structure and select their best work from each lesson. They write descriptions explaining the technical skills demonstrated in each piece.

Elaborate (10 min)

Teacher Create your complete graphics portfolio in GIMP. Include a cover page, table of contents, and at least eight pieces of work with descriptions. Ensure consistent formatting throughout. Export as both GIMP XCF files and a PDF portfolio document for sharing.

Students students compile professional portfolios demonstrating their complete skill set. They practice file management and presentation skills.

Evaluate (5 min)

Teacher Present your portfolio to the class. Explain your design choices for each piece and what skills you demonstrated. Receive feedback from classmates on both the work quality and presentation effectiveness.

Students students present confidently and articulate their creative and technical decisions. They learn from peer feedback and recognize their growth.

Cross-Curricular Connections

Mathematics: Calculate the average quality score of your portfolio pieces based on self-assessment and peer feedback using a one to five scale.
Science: Explain how digital portfolios are stored on servers and accessed through web browsers using cloud hosting services.
Language: Write a reflective essay about your graphics learning journey describing challenges overcome and skills gained.

Digital Citizen Reminder: Only include work you created yourself in your portfolio. Properly credit any resources, templates, or images you used. Be honest about your skill level and learning process.

Resources Needed:
- All previous lesson files
- Portfolio template
- PDF export settings
- Reflection prompt sheet
- Presentation area

Differentiation

Approaching: Provide a portfolio template with pre-made pages where students insert their work and write descriptions
On Level: Students create complete portfolios following standard organization and documentation requirements
Advanced: Advanced students create an online portfolio website using free tools to showcase their work professionally
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Portfolio rubric evaluating completeness, quality, organization, and reflection
CT Skill Assessed: Curation selecting and organizing best work to demonstrate competence

Dormitory — Retrieval (Paper)

Think about which graphics project from this chapter you are most proud of. Write three sentences explaining why it is your best work.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Chapter 3 — Multimedia

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Engage (5 min)

Teacher Think about watching a movie. You see pictures moving, hear music and dialogue, read subtitles, and sometimes interact with the story through games. How many different types of media are working together in this single experience?

Students students count visual, audio, text, and interactive elements. They realize that movies combine many media forms into one experience.

Explore (10 min)

Teacher Examine these examples on the board. A textbook page uses text and images. A radio program uses audio only. A television show uses video and audio. A website uses text, images, audio, and video. Which of these is multimedia and why?

Students students identify that websites and television shows are multimedia because they combine multiple media types while radio uses only audio.

Explain (10 min)

Teacher Multimedia means multiple forms of media working together. The five elements are text for information, graphics for visual appeal, audio for sound, video for moving images, and animation for motion effects. Multimedia can be linear like a movie that plays from start to finish or interactive like a game where users control the experience. Multimedia applications include education, entertainment, advertising, and communication.

Students students create a multimedia elements chart with examples of each type. They categorize different media experiences as linear or interactive.

Elaborate (10 min)

Teacher Think about your favorite mobile game or educational app. List all the multimedia elements it uses. How do text, graphics, audio, and video work together to create the experience? Present your analysis to the class.

Students students analyze apps identifying text labels, graphic characters, sound effects, background music, and animated transitions. They see how elements combine for engagement.

Evaluate (5 min)

Teacher Classify each example as multimedia or single media. Online newspaper. Podcast. Interactive museum kiosk. Television advertisement. Digital textbook with videos.

Students students correctly identify that interactive kiosk and digital textbook are multimedia while podcast uses only audio. They explain their reasoning.

Cross-Curricular Connections

Mathematics: Calculate the number of multimedia elements on a typical educational website by counting text blocks, images, videos, and audio players.
Science: Explain how the human brain processes visual information sixty thousand times faster than text and how multimedia leverages this for learning.
Language: Write a proposal for a multimedia presentation about Karnataka culture that includes text, images, audio, and video elements.

Digital Citizen Reminder: Multimedia content can be manipulated to spread misinformation. Always verify multimedia sources before believing or sharing them. Check facts across multiple reliable sources.

Resources Needed:
- Multimedia examples
- Textbook
- Computer with internet
- Projector
- Sample multimedia files

Differentiation

Approaching: Provide picture cards showing different media types for sorting and categorization activities
On Level: Students complete standard multimedia identification and classification exercises
Advanced: Advanced students research augmented reality and virtual reality as emerging multimedia forms
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Multimedia elements identification and classification worksheet
CT Skill Assessed: Pattern recognition identifying different data types in complex media

Dormitory — Retrieval (Paper)

List all multimedia elements you used today. Include videos watched, music heard, text read, and images seen on your phone or computer.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Have you ever recorded a voice message on your phone and wished you could remove the um and ah sounds? Or add background music to make it sound professional? What tools would you need for this?

Students students think about editing voice messages and adding music. They realize audio editing can transform rough recordings into polished content.

Explore (10 min)

Teacher Launch Audacity on the school computer. Explore the interface. Find the record button, play button, stop button, and the timeline where audio waves appear. Try recording five seconds of silence and then play it back. Observe the flat line representing silence.

Students students discover Audacity has transport controls like a tape recorder. They see that silence appears as a flat line and sound appears as wavy lines.

Explain (10 min)

Teacher Audacity is free audio editing software. The interface has the transport controls at top left for record, play, stop, and rewind. The timeline shows audio as waveforms from left to right. The track control panel on the left has mute and solo buttons. You can zoom in and out to see detail. Key operations include cut to remove selected audio, copy to duplicate, paste to insert, and trim to keep only selected audio. Export formats include MP3 for compressed sharing and WAV for high quality.

Students students practice recording and basic editing. They learn to select audio by clicking and dragging, then cut, copy, and paste sections.

Elaborate (10 min)

Teacher Record yourself reading a paragraph from your textbook. Edit out any mistakes or long pauses. Add a two-second silence at the beginning for a professional start. Export the edited audio as an MP3 file with your name in the filename.

Students students complete a practical editing project. They practice selection precision and learn to maintain audio quality during editing.

Evaluate (5 min)

Teacher Complete these tasks. Record ten seconds of audio. Cut the middle three seconds. Copy and paste the first two seconds to the end. Export as MP3. Explain each step you performed.

Students students demonstrate competence in basic Audacity operations. They articulate their editing process clearly.

Cross-Curricular Connections

Mathematics: Calculate the file size of a three-minute audio recording at CD quality of forty-four point one kilohertz, sixteen-bit stereo.
Science: Explain how microphones convert sound waves into electrical signals that computers store as digital audio samples.
Language: Write a script for a two-minute educational audio recording about a science topic you are studying.

Digital Citizen Reminder: Only record audio with permission from the people being recorded. Respect copyright when adding music to your projects. Use royalty-free music from legitimate sources.

Resources Needed:
- Computers with Audacity installed
- Microphones
- Headphones
- Audio sample files
- Textbook

Differentiation

Approaching: Provide step-by-step picture guides for each Audacity operation with numbered screenshots
On Level: Students complete standard recording and editing exercises following verbal instructions
Advanced: Advanced students explore noise removal and equalization for professional audio quality
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Audio editing practical test with recording and export task
CT Skill Assessed: Sequential processing executing steps in correct order to achieve desired outcome

Dormitory — Retrieval (Paper)

Record a thirty-second voice message describing your day at KREIS school. Play it back and note any sounds you would like to remove.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Think about the last audio or video you watched online. Was the voice clear and easy to understand? What made it sound professional versus amateur? What environment and equipment do professional podcasters use?

Students students recall clear professional audio versus echoey phone recordings. They identify quiet environment and good microphone as key factors.

Explore (10 min)

Teacher Test different recording positions with the school microphone. Record the same sentence from ten centimeters, thirty centimeters, and one meter away. Listen to each recording and compare clarity, volume, and background noise.

Students students discover that distance affects audio quality significantly. Close recordings have more bass and less room echo while distant recordings sound thin and echoey.

Explain (10 min)

Teacher Good audio recording requires attention to environment, equipment, and settings. Environment should be quiet with soft surfaces to reduce echo. Position the microphone fifteen to twenty centimeters from your mouth slightly to the side to avoid plosive pops. Sample rate of forty-four point one kilohertz is standard for music. Bit depth of sixteen bits provides good dynamic range. Record at moderate levels to avoid distortion and leave headroom for editing.

Students students practice proper microphone technique. They learn to check recording levels and adjust gain for optimal sound without clipping.

Elaborate (10 min)

Teacher Set up a recording station in a quiet corner of the classroom. Record a one-minute narration about your favorite place in the school. Focus on speaking clearly, maintaining consistent distance from the microphone, and keeping background noise minimal.

Students students apply recording techniques in a practical setting. They learn that preparation and environment are as important as equipment.

Evaluate (5 min)

Teacher Record the same paragraph three times with different microphone positions. Identify which recording sounds best and explain why. Rate your recording on clarity, volume consistency, and background noise.

Students students develop critical listening skills. They can identify quality differences and explain the technical reasons behind them.

Cross-Curricular Connections

Mathematics: Calculate the data rate of a recording at forty-four point one kilohertz, sixteen-bit, stereo in kilobits per second.
Science: Explain how sound travels as waves through air and how microphone diaphragms convert these waves to electrical signals.
Language: Write a recording script with introduction, main content, and conclusion for a five-minute educational podcast episode.

Digital Citizen Reminder: Always get permission before recording others. Inform people when they are being recorded. Never record private conversations without consent.

Resources Needed:

- Microphones
- Quiet recording space
- Pop filter if available
- Audacity software
- Recording checklist

Differentiation

Approaching: Provide a recording checklist with pictures showing correct microphone position and settings
On Level: Students complete standard recording exercises following environmental and technical guidelines
Advanced: Advanced students explore stereo recording and binaural audio techniques for immersive sound
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Recording quality assessment using checklist criteria
CT Skill Assessed: Environmental optimization adjusting conditions for best possible output

Dormitory — Retrieval (Paper)

Find the quietest spot in your hostel. Record a voice message there. Then record the same message in the corridor. Compare which sounds better and why.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Listen to this recording. It has background hiss, sounds muffled, and the volume goes up and down. Can you imagine using tools to fix each of these problems? What would a professional recording studio do?

Students students identify three common audio problems and wonder if software can fix them. They learn that effects can dramatically improve audio quality.

Explore (10 min)

Teacher Open a noisy recording in Audacity. Select a quiet section with only background noise. Go to Effect menu and choose Noise Reduction. Click Get Noise Profile. Now select the entire track and apply Noise Reduction. Listen to the improvement. Try different reduction levels.

Students students experience that noise removal works by identifying background noise pattern and subtracting it from the recording. Too much removal creates artifacts.

Explain (10 min)

Teacher Audio effects improve sound quality. Noise reduction removes background hiss and hum by sampling noise and subtracting it. Equalization adjusts bass, mid, and treble frequencies to brighten or warm the sound. Compression reduces the difference between loud and quiet parts for consistent volume. Reverb adds space and depth making recordings sound like they were made in a larger room. Each effect should be used subtly to enhance without distorting.

Students students practice applying effects to sample recordings. They learn that less is more with audio effects and 过度 processing sounds unnatural.

Elaborate (10 min)

Teacher Take your voice recording from the previous lesson. Apply noise reduction to remove background noise. Use equalization to boost clarity in the voice frequency range. Apply light compression for consistent volume. Export the processed version and compare with the original.

Students students apply a complete effects chain transforming amateur recordings into professional-sounding audio. They document their settings for each effect.

Evaluate (5 min)

Teacher Apply noise reduction to this recording and explain your settings. Use equalization to make this voice sound warmer. Apply compression to even out the volume levels. Evaluate whether each effect improved or degraded the audio.

Students students demonstrate practical effects knowledge. They can identify when effects help and when they cause problems.

Cross-Curricular Connections

Mathematics: Calculate the frequency range of human hearing from twenty hertz to twenty thousand hertz. What percentage of this range does a typical voice occupy?
Science: Explain how equalization works by dividing audio into frequency bands and adjusting the volume of each band independently.
Language: Write a tutorial explaining three audio effects to someone who has never used audio editing software.

Digital Citizen Reminder: Audio effects can be used to deceive by making fake recordings sound authentic. Be aware that voice recordings can be manipulated and verify important audio through multiple sources.

Resources Needed:
- Audacity with effects menu
- Noisy sample recordings
- Effect settings guide
- Headphones for critical listening
- Textbook

Differentiation

Approaching: Provide pre-configured effect presets with recommended settings for common audio problems
On Level: Students apply standard effects following step-by-step instructions with specific parameter values
Advanced: Advanced students explore multiband compression and dynamic EQ for broadcast-quality processing
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Before and after audio comparison with effects documentation
CT Skill Assessed: Iterative refinement applying successive improvements to achieve optimal quality

Dormitory — Retrieval (Paper)

Listen to a song on your phone and focus on the vocals. Does the singer sound like they are in a small room or a large hall? What audio effect might create that feeling?

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

> **Teacher** Watch this raw footage from a school event. It has long boring parts, shaky camera movements, and random scenes in wrong order. How would you transform this into an exciting three-minute highlight video?

> **Students** students identify problems in raw footage and brainstorm editing solutions. They realize that editing transforms raw material into compelling stories.

Explore (10 min)

Teacher Open a video editor on the school computer. Import two video clips. Notice how they appear in the media library. Drag them onto the timeline. Play the timeline to see them play in sequence. Try trimming the end of the first clip by dragging its edge.

Students students experience the basic workflow of importing, arranging, and trimming. They find that the timeline represents the sequence of their video.

Explain (10 min)

Teacher Video editing arranges clips to tell a story. Import brings media files into the project. The timeline shows clips from left to right in playback order. Trim removes unwanted parts from the beginning or end of a clip. Split cuts a clip into two parts at a specific point. Transitions like fade and dissolve smooth the change between clips. Titles add text information like names and locations. Always save your project frequently to avoid losing work.

Students students practice basic editing operations. They learn to use keyboard shortcuts for faster editing and to save frequently.

Elaborate (10 min)

Teacher Create a one-minute video about a day at KREIS school. Import at least five video clips. Trim each to keep only the best moments. Arrange them in chronological order. Add a title at the beginning and credits at the end. Add simple fade transitions between clips.

Students students complete a practical video project. They practice making editorial decisions about what to keep and what to cut.

Evaluate (5 min)

Teacher Trim this thirty-second clip to remove the first five seconds and last three seconds. Split this clip at the point where the action changes. Add a fade transition between these two clips. Explain each editing decision you made.

Students students demonstrate basic editing competence. They can articulate their editorial reasoning.

Cross-Curricular Connections

Mathematics: Calculate the total duration of a video project with eight clips averaging four seconds each plus three transitions of one second each.
Science: Explain how video frames work at twenty-four or thirty frames per second to create the illusion of smooth motion.
Language: Write a storyboard plan for a two-minute school promotional video listing each scene, duration, and type of shot.

Digital Citizen Reminder: Only use video footage you have created or have permission to use. Respect copyright when adding music. Give credit to anyone who appears in your video.

Resources Needed:

- Video editing software
- Sample video clips
- Microphone for narration
- Projector for demonstration
- Textbook

Differentiation

Approaching: Provide pre-made video clips and a template timeline for simpler editing exercises
On Level: Students create standard video projects following editing guidelines
Advanced: Advanced students explore multi-track editing with separate audio and video tracks
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Video editing project evaluation using editing criteria rubric
CT Skill Assessed: Sequential narrative construction organizing visual elements into coherent story

Dormitory — Retrieval (Paper)

Watch a short video on YouTube. Count how many different shots appear in the first thirty seconds. Notice how quickly professional videos change scenes.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

> **Teacher** Watch these two video clips. The first cuts abruptly from one scene to the next. The second uses a smooth dissolve to blend one scene into the other. Which feels more natural? When would you use a hard cut versus a soft dissolve?

> **Students** students notice that abrupt cuts feel jarring while dissolves feel gentle. They begin to understand that transition choice affects emotional response.

Explore (10 min)

Teacher Open your video project from the last lesson. Try adding different transitions between clips. Use a cut, then a fade to black, then a dissolve, then a wipe. Play each one and note how it changes the feeling of the video.

Students students experience that each transition creates a different mood. Fade to black feels final. Dissolve feels dreamlike. Wipe feels dynamic.

Explain (10 min)

Teacher Transitions connect scenes and control pacing. Cut is instant and most common used for same-time same-place scene changes. Dissolve blends two clips together over one to two seconds for gentle transitions. Fade to black gradually darkens to black screen indicating time passage or scene end. Wipe slides one clip off while another slides on for dynamic transitions. Key rule is to match transition to content. Do not use more than two different transition types in one video.

Students students practice selecting transitions based on content analysis. They learn that simple is usually better than flashy.

Elaborate (10 min)

Teacher Edit your school day video using only appropriate transitions. Use cuts between same-location scenes. Use dissolves for time passage. Use fade to black at the end. Write a transition guide explaining why you chose each transition at each point.

Students students demonstrate understanding of transition purpose. They can justify each transition choice with content-based reasoning.

Evaluate (5 min)

Teacher Identify the transition type used between each pair of clips in this sample video. Rate whether each transition is appropriate for the content. Suggest better alternatives where transitions do not match the mood.

Students students develop critical evaluation skills for transition effectiveness. They can identify mismatches and propose improvements.

Cross-Curricular Connections

Mathematics: Calculate the total transition time if a two-minute video has six transitions each lasting one point five seconds.
Science: Explain how the human eye perceives smooth motion at twenty-four frames per second and how transitions use frame blending.
Language: Write a film analysis paragraph describing how transitions contribute to the mood of a movie scene you watched recently.

Digital Citizen Reminder: Transitions should serve the story not distract from it. Overuse of fancy transitions can make educational videos look unprofessional and reduce learning effectiveness.

Resources Needed:
- Video editing software
- Transition examples
- Sample video clips
- Transition guide chart
- Textbook

Differentiation

Approaching: Provide a transition matching activity where students match transition types to appropriate content situations
On Level: Students apply standard transitions following editing best practices guidelines
Advanced: Advanced students explore custom transitions using masking and keyframe animation
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Transition analysis worksheet and video editing exercise
CT Skill Assessed: Contextual decision making choosing appropriate tools based on content requirements

Dormitory — Retrieval (Paper)

Watch a movie or TV show tonight and pay attention to transitions between scenes. Write down three different transitions you notice and why the director might have chosen each one.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Watch this bouncing ball animation. Now watch this one. Both show a ball bouncing but one looks realistic and alive while the other looks stiff and robotic. What makes the difference between smooth and jerky animation?

Students students notice that realistic animation has timing variation and squash-stretch while robotic animation has uniform motion. They want to learn the secrets of smooth animation.

Explore (10 min)

Teacher Open a simple animation tool. Create a ball at the top of the screen. Set a keyframe. Move the ball to the bottom and set another keyframe. Play the animation. Now add a third keyframe where the ball squashes at the bottom. Notice how the motion changes.

Students students experience that keyframes define important positions and the computer fills in the motion between them. Squash and stretch adds realism.

Explain (10 min)

Teacher Animation principles guide believable motion. Squash and stretch gives weight and flexibility. Anticipation prepares the viewer for an action. Staging directs attention to the most important element. Straight ahead action creates fluid natural motion. Follow through and overlapping action show that different parts move at different speeds. Slow in and slow out makes motion feel natural by starting and ending slowly. Arcs follow natural curved paths.

Students students create a bouncing ball animation applying squash, stretch, and slow in slow out principles. They observe that following principles creates more natural motion.

Elaborate (10 min)

Teacher Create a simple animation of a character walking across the screen. Apply at least four animation principles. Use anticipation before each step. Add follow through on the arms. Use slow in and slow out for natural movement. Stage the character clearly against the background.

Students students apply multiple principles to create complex believable motion. They learn that combining principles produces better results than using one alone.

Evaluate (5 min)

Teacher Identify which animation principles are used in this sample animation. Create a bouncing ball animation that demonstrates squash, stretch, and timing. Explain why each principle makes the animation look more realistic.

Students students demonstrate understanding of animation theory and practical application. They can articulate how principles affect visual quality.

Cross-Curricular Connections

Mathematics: Calculate the number of frames needed for a bouncing ball animation at thirty frames per second if each bounce takes half a second.
Science: Explain how persistence of vision allows the human eye to perceive individual frames as continuous motion at speeds above twelve frames per second.
Language: Write a storyboard for a ten-second animation showing a character performing a simple action with labeled animation principles.

Digital Citizen Reminder: Animation should be used to enhance learning not distract from it. Avoid excessive animation in educational materials that can overwhelm viewers and reduce comprehension.

Resources Needed:
- Animation software
- Principles reference chart
- Storyboard templates
- Sample animations
- Textbook

Differentiation

Approaching: Provide pre-made keyframe templates where students adjust timing values to see principle effects
On Level: Students create standard animation exercises following principle guidelines
Advanced: Advanced students explore character rigging and walk cycle animation with secondary motion
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Animation principles identification and application exercise
CT Skill Assessed: Physics simulation understanding how real-world forces affect motion in digital environments

Dormitory — Retrieval (Paper)

Watch a short animation on your phone. Pause at different points and try to identify where keyframes might have been placed to create the motion.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Think about the worst presentation you have ever seen. What made it bad? Now think about the best one. What made it engaging and memorable? How would you create a presentation that people actually enjoy watching?

Students students recall presentations with too much text and boring layouts versus ones with great images and storytelling. They want to learn effective presentation design.

Explore (10 min)

Teacher Open presentation software and explore the available themes. Notice how each theme changes fonts, colors, and layout. Try the slide master to see how it controls the design of all slides. Create a title slide with the school name and your name.

Students students discover that themes provide consistent professional design. They learn that slide master controls global design elements.

Explain (10 min)

Teacher Effective presentations follow the ten-twenty-thirty rule. Ten slides maximum for a twenty-minute presentation with thirty point minimum font. Each slide should have one main idea. Use images more than text. Use bullet points with maximum six items per slide. Maintain consistent fonts and colors throughout. Use slide transitions sparingly. Include speaker notes for your talking points not on the slide.

Students students redesign a bad presentation applying these principles. They transform text-heavy slides into visual impactful slides.

Elaborate (10 min)

Teacher Create a five-slide presentation about your favorite topic. Apply the ten-twenty-thirty rule adapted for five slides. Use consistent theme throughout. Include at least one image per slide. Add speaker notes for each slide explaining what you would say during the presentation.

Students students create complete professional presentations. They practice balancing visual content with verbal explanation.

Evaluate (5 min)

Teacher Evaluate this presentation against the ten-twenty-thirty rule. Identify three slides that have too much text. Suggest image alternatives for text-heavy content. Present your own creation to a partner and get feedback.

Students students apply evaluation criteria to existing work and receive peer feedback on their own presentations. They iterate to improve.

Cross-Curricular Connections

Mathematics: Calculate the ideal font size for a presentation viewed from ten meters using the thirty point rule and viewing distance guidelines.
Science: Explain how presentation software uses vector graphics for text and shapes that scale without losing quality at any resolution.
Language: Write a persuasive speech to accompany a five-slide presentation about why KREIS schools need a new library.

Digital Citizen Reminder: Presentation software can be used to present misleading data through manipulated charts and graphs. Always represent data accurately and cite your sources.

Resources Needed:
- Presentation software
- Sample presentations
- Design principles guide
- Image resources
- Textbook

Differentiation

Approaching: Provide presentation templates with pre-designed slides where students add content to specified areas
On Level: Students create standard presentations following design principles and guidelines
Advanced: Advanced students create interactive presentations with hyperlinks and embedded quizzes
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Presentation design rubric evaluating visual design and content organization
CT Skill Assessed: Information design structuring content for maximum audience comprehension and engagement

Dormitory — Retrieval (Paper)

Create three slides about your dormitory on your phone using any presentation app. Use one image per slide and keep text minimal.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher What if a presentation could respond to audience questions and show different content based on what people want to learn? Like a choose your own adventure story but for presentations. How would you build something like this?

Students students imagine presentations that branch into different paths. They get excited about creating interactive experiences.

Explore (10 min)

Teacher Open a presentation and create two different content paths. Add a button on slide one that links to slide three and another button that links to slide five. Click each button and see how the presentation jumps to different slides.

Students students discover that hyperlinks can jump to any slide creating branching paths. They realize this enables interactive storytelling.

Explain (10 min)

Teacher Interactive presentations break the linear slide-by-slide format. Hyperlinks connect text or objects to specific slides creating navigation options. Action buttons provide clickable areas for audience choices. Embedded quizzes test understanding during the presentation. Branching scenarios let audiences choose their learning path. Tools like Google Slides and PowerPoint support these features. Design interactive elements to be large enough to click and clearly labeled.

Students students add interactivity to their presentations. They create quiz slides with answer feedback and navigation menus for topic selection.

Elaborate (10 min)

Teacher Create an interactive quiz presentation about a chapter from your textbook. Include a title slide, question slides with multiple choice options, correct answer feedback slides, and a final score slide. Each answer choice links to the appropriate feedback.

Students students build complete interactive quiz experiences. They practice designing clear navigation and meaningful feedback for each answer.

Evaluate (5 min)

Teacher Test your interactive presentation with a classmate. Observe whether they can navigate without instructions. Identify any confusing links or unclear buttons. Revise based on feedback to improve usability.

Students students test their creations with real users. They learn that good design is invisible and confusing design frustrates users.

Cross-Curricular Connections

Mathematics: Calculate the number of possible paths through a five-question quiz where each question has three answer choices.
Science: Explain how hyperlinks work using URL addresses that tell the browser or application where to jump when clicked.
Language: Write a flowchart showing all possible paths through your interactive quiz from start to finish.

Digital Citizen Reminder: Interactive presentations should respect audience autonomy. Do not force audiences through predetermined paths without offering choices. Allow people to navigate at their own pace.

Resources Needed:
- Presentation software
- Quiz question bank
- Flowchart templates
- Navigation icons
- Textbook

Differentiation

Approaching: Provide a pre-made quiz template where students replace questions and answers with their own content
On Level: Students create standard interactive presentations with branching navigation
Advanced: Advanced students create adaptive presentations using macros and conditional logic
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Interactive presentation functionality test and user feedback evaluation
CT Skill Assessed: User experience design creating intuitive interfaces for audience interaction

Dormitory — Retrieval (Paper)

Think of a quiz about your school. Write five multiple choice questions with three answer options each and indicate which option is correct.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Tell me about a story that made you cry, laugh, or feel excited. What elements did the storyteller use? Was it the words, the pictures, the music, or the way they told it? How can we use multimedia to create powerful stories?

Students students recall emotional stories and identify elements that created feelings. They realize multimedia can amplify storytelling power.

Explore (10 min)

Teacher Watch three different digital stories about the same topic. One uses only text. One uses text with images. One uses text, images, music, and narration. Rate each for emotional impact and engagement. What made the third version most powerful?

Students students find that adding multimedia layers increases engagement and emotional connection. They notice music especially affects mood.

Explain (10 min)

Teacher Digital storytelling uses multimedia to tell narratives. Structure follows the story arc. Beginning introduces characters and setting. Rising action builds tension and conflict. Climax is the turning point. Falling action resolves the conflict. Conclusion provides closure. Multimedia elements reinforce the narrative. Images show what words describe. Music sets the emotional tone. Narration personalizes the story. Text provides context and captions.

Students students plan a digital story using the story arc structure. They select multimedia elements that support each story section emotionally.

Elaborate (10 min)

Teacher Create a two-minute digital story about a personal experience at KREIS school. Write a script following the story arc. Select or create images that match each story moment. Record narration with appropriate emotional tone. Add background music that enhances the mood. Combine all elements in a video editor.

Students students produce complete digital stories integrating all multimedia skills. They practice storytelling craft alongside technical skills.

Evaluate (5 min)

Teacher Present your digital story to the class. After presenting, answer questions about your creative choices. What image was most powerful and why? How did music affect the mood? What would you do differently next time?

Students students present and defend their creative choices. They receive feedback on both technical execution and storytelling effectiveness.

Cross-Curricular Connections

Mathematics: Calculate the ideal narration speed for clear storytelling at one hundred fifty words per minute for a two-minute story.
Science: Explain how the human brain processes visual information and emotional music simultaneously to create stronger memory formation.
Language: Write a detailed story script with scene descriptions, narration text, and music cues for a three-minute digital story.

Digital Citizen Reminder: Digital stories about real people should be told with their permission and dignity. Do not exploit personal experiences for entertainment without consent. Represent all people respectfully.

Resources Needed:
- Storytelling structure guide
- Image resources
- Music library
- Narration recording setup
- Video editor
- Textbook

Differentiation

Approaching: Provide story templates with pre-written story arcs where students add personal details and select multimedia elements
On Level: Students create standard digital stories following the story arc structure
Advanced: Advanced students create interactive digital stories with audience choices affecting the narrative outcome
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Digital story evaluation rubric with storytelling and technical criteria
CT Skill Assessed: Narrative construction combining multiple media types into coherent emotional experience

Dormitory — Retrieval (Paper)

Think about your first day at KREIS school. Write the story in three sentences. What image would you choose to represent that day? What music would play in the background?

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

> **Teacher** Do you listen to podcasts? What makes a podcast different from a radio show or an audio recording? What elements do your favorite podcasts include that keep you listening?

> **Students** students discuss podcast elements like regular episodes, host personality, segments, and production quality. They want to create their own.

Explore (10 min)

> **Teacher** Listen to this sample podcast episode. Note the intro music, host introduction, main content, interviews, and outro. Create a timeline showing each segment and its duration. How long is each part?

> **Students** students discover that podcasts have structured segments with different purposes. They find that intros and outros create professional framing.

Explain (10 min)

> **Teacher** A podcast is a regularly released audio program with episodes. Planning includes choosing a topic, writing a script outline, identifying segments, and scheduling release dates. Episode structure typically includes intro music and greeting, segment one with main topic, segment two with interview or discussion, listener questions or feedback, and outro with next episode preview. Recording requires quiet environment and consistent audio levels. Editing removes mistakes and tightens pacing.

> **Students** students create a podcast planning document with topic, segments, and timing. They write a script outline for a pilot episode.

Elaborate (10 min)

> **Teacher** Record a ten-minute podcast episode about an educational topic. Include a thirty-second intro with music, a main segment of five minutes, a guest interview segment of three minutes recorded with a classmate, and a one-minute outro with music. Edit for clarity and add transitions between segments.

> **Students** students produce complete podcast episodes. They practice interviewing, segment transitions, and maintaining consistent audio quality throughout.

Evaluate (5 min)

Teacher Listen to your podcast and evaluate it against these criteria. Is the intro engaging? Are segments clearly separated? Is the audio quality consistent? Does the outro encourage listeners to return? What would improve the next episode?

Students students critically evaluate their own productions. They identify strengths and specific areas for improvement in future episodes.

Cross-Curricular Connections

Mathematics: Calculate the total episode length by adding segment durations and transition times. If intro is thirty seconds, main is five minutes, interview is three minutes, and outro is one minute, what is the total?
Science: Explain how podcast distribution works through RSS feeds that automatically deliver new episodes to subscribers through podcast apps.
Language: Write a podcast episode guide including topic research, interview questions, and discussion points for a ten-minute educational episode.

Digital Citizen Reminder: Podcast hosts have responsibility for accurate information. Verify facts before broadcasting. Give credit to sources and guests. Create content that is respectful and inclusive of all listeners.

Resources Needed:

- Microphones
- Quiet recording space
- Audacity software
- Royalty-free music
- Podcast planning template
- Textbook

Differentiation

Approaching: Provide podcast script templates with pre-written segments where students add their own content
On Level: Students create standard podcast episodes following structure guidelines
Advanced: Advanced students create podcast series with consistent branding, episode numbering, and regular release schedule
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Podcast episode evaluation using production quality rubric
CT Skill Assessed: Content production creating engaging audio content through planning and technical execution

Dormitory — Retrieval (Paper)

Listen to a five-minute podcast episode tonight. Write down the intro, main topic, and outro. Note the total duration of each segment.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher You have learned audio editing, video editing, animation, presentations, digital storytelling, and podcasting. Now you will combine all these skills into one comprehensive multimedia project. What topic would you like to showcase using all these media forms?

Students students brainstorm topics that could be expressed through multiple media. They excitedly plan projects that demonstrate their complete skill set.

Explore (10 min)

Teacher Examine these award-winning student multimedia projects. One is about school heritage using video, audio interviews, and photos. Another is about science experiments using animation, narration, and data visualization. Define what makes each project effective.

Students students identify that successful projects combine media types purposefully. They notice clear planning, consistent theme, and professional polish.

Explain (10 min)

Teacher A multimedia project requires careful planning. Define your topic and audience. Create a project proposal with objectives and media plan. Storyboard each section identifying media type and content. Gather or create all media assets. Build the project in stages. Test with audience and revise. Document your process and choices. Present professionally with introduction and Q and A session.

Students students create project proposals and storyboards. They plan their work in stages with clear milestones and deadlines.

Elaborate (10 min)

Teacher Execute your multimedia project plan. Create at least four different media elements. Combine them into a final product. Write documentation explaining your creative choices and technical process. Prepare a five-minute presentation about your project for the class.

Students students produce complete multimedia projects demonstrating integrated skills. They practice project management and documentation.

Evaluate (5 min)

Teacher Present your multimedia project to the class. Explain your media choices, creative decisions, and technical challenges. Receive feedback from classmates using a structured evaluation form. Reflect on what you would do differently next time.

Students students present confidently and respond to questions professionally. They learn from peer feedback and self-reflection.

Cross-Curricular Connections

Mathematics: Calculate the total production time for your project by tracking hours spent on planning, creation, editing, and presentation preparation.
Science: Explain how multimedia projects require understanding of human perception across visual, auditory, and cognitive channels simultaneously.
Language: Write a project reflection essay discussing what you learned, what challenged you, and how you would improve your approach next time.

Digital Citizen Reminder: Multimedia projects should respect intellectual property. Credit all sources, get permission for others work, and create original content whenever possible. Be proud of your own authentic work.

Resources Needed:
- All previous lesson tools
- Project planning templates
- Documentation guidelines
- Presentation space
- Evaluation forms
- Textbook

Differentiation

Approaching: Provide project templates with pre-defined structure where students fill in content for each media section
On Level: Students create complete multimedia projects following standard planning and execution guidelines
Advanced: Advanced students create multi-chapter multimedia projects with cross-references and interactive navigation
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Comprehensive project rubric with planning, execution, documentation, and presentation criteria
CT Skill Assessed: Project integration combining multiple skills into a unified professional-quality product

Dormitory — Retrieval (Paper)

Write a one-page project plan for a multimedia story about your family. List four media elements you would include and explain why each is necessary.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Chapter 4 — Data Handling

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Engage (5 min)

Teacher Imagine you need to track the marks of all fifty students in your class for five subjects. Doing this on paper would take hours and mistakes would be hard to fix. What if there was a tool that could calculate totals, averages, and grades automatically?

Students students realize that spreadsheets save time and reduce calculation errors. They imagine automatic grade calculation versus manual addition.

Explore (10 min)

Teacher Open the spreadsheet application. Click on different cells and notice the cell address in the name box like A1, B2, C3. Type your name in cell A1. Type a number in B1. Try using arrow keys to move between cells. Notice the formula bar shows the cell content.

Students students discover that cells are identified by column letter and row number. They find that each cell can hold text, numbers, or formulas.

Explain (10 min)

Teacher A spreadsheet is a grid of cells organized in rows numbered one, two, three and columns labeled A, B, C. Each cell holds one piece of data. The intersection of column C and row five is cell C5. Spreadsheets are used for data entry, calculations, analysis, and visualization. Key features include formulas that calculate automatically, formatting that makes data readable, sorting that organizes data, and charts that visualize patterns.

Students students practice navigating and entering data. They create a simple table with student names in column A and marks in column B.

Elaborate (10 min)

Teacher Create a class attendance tracker for the week. Enter student names in column A. Enter days Monday through Friday in row one. Mark present with P and absent with A. Format the header row with bold text and background color.

Students students apply spreadsheet basics to create a practical document. They learn formatting improves readability of data tables.

Evaluate (5 min)

Teacher Answer these questions. What is a cell address? How do you move between cells? What are three types of data you can enter in a cell? Create a simple table with five rows and three columns.

Students students demonstrate understanding of spreadsheet fundamentals. They navigate confidently and enter data correctly.

Cross-Curricular Connections

Mathematics: Calculate the total number of cells in a spreadsheet with one hundred rows and twenty-six columns.
Science: Explain how spreadsheet software stores data in memory using arrays and how cell references work in calculation.
Language: Write a report comparing paper registers versus digital spreadsheets for tracking student information at KREIS school.

Digital Citizen Reminder: Spreadsheet data about students is sensitive personal information. Protect files with passwords and do not share student marks with unauthorized people.

Resources Needed:
- Spreadsheet software installed
- Practice data sheets
- Formatting guide
- Textbook
- Projector for demonstration

Differentiation

Approaching: Provide a pre-made spreadsheet template with labels where students enter their own data in specified cells
On Level: Students create spreadsheets from scratch following verbal instructions
Advanced: Advanced students explore named ranges and multiple worksheets within a workbook
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Spreadsheet navigation and data entry task
CT Skill Assessed: Data organization structuring information in grid-based systems for efficient access

Dormitory — Retrieval (Paper)

Create a simple spreadsheet on your phone or computer listing five items you need to buy from the store with their estimated prices in a column next to them.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Look at this messy spreadsheet. Numbers are left-aligned, text is right-aligned, there are no borders, and everything looks the same. Now look at this formatted version. Same data but organized with colors, borders, and proper alignment. Which one would you rather use?

Students students immediately prefer the formatted version. They realize that formatting transforms raw data into readable information.

Explore (10 min)

Teacher Enter a list of numbers in column A. Notice they right-align by default. Enter text in column B. Notice it left-aligns by default. Select all cells with numbers and change alignment to center. Add borders around the data. Apply a background color to the header row.

Students students discover that spreadsheets automatically detect data types and align accordingly. They learn that formatting makes data presentation professional.

Explain (10 min)

Teacher Data types determine how information is stored and displayed. Text aligns left by default. Numbers align right. Dates are stored as serial numbers but displayed as dates. Formatting includes alignment left center right, borders thin thick colored, fill colors for highlighting, font bold italic underline, and number format decimal currency percentage. The fill handle at the bottom-right corner of a selected cell copies patterns to adjacent cells automatically.

Students students practice formatting a sample data table. They use auto-fill to create number sequences and day lists efficiently.

Elaborate (10 min)

Teacher Create a formatted monthly expense tracker. Enter expense categories in column A, amounts in column B, and dates in column C. Format amounts as currency with rupee symbol. Format dates consistently. Add borders around all cells. Use fill colors to separate categories. Apply bold formatting to headers.

Students students create professionally formatted spreadsheets. They practice combining multiple formatting features for clear data presentation.

Evaluate (5 min)

Teacher Format this raw data table with proper alignment, borders, colors, and number formatting. Use auto-fill to extend this number sequence from one to twenty. Explain the difference between formatting cells and changing cell content.

Students students demonstrate formatting competence. They understand that formatting changes appearance without changing underlying data.

Cross-Curricular Connections

Mathematics: Calculate the number of different formatting combinations possible with three alignment options, five border styles, and ten fill colors.
Science: Explain how spreadsheet software stores formatted data differently from raw data, keeping display properties separate from values.
Language: Write a formatting guide for younger students explaining when to use each alignment option and color convention.

Digital Citizen Reminder: Color coding should be accessible to color-blind users. Do not rely solely on color to convey meaning. Use patterns or labels in addition to colors for inclusive data presentation.

Resources Needed:
- Spreadsheet software
- Raw data samples
- Formatting reference sheet
- Color accessibility guide
- Textbook

Differentiation

Approaching: Provide pre-formatted templates where students practice specific formatting tasks on provided data
On Level: Students complete standard formatting exercises following guidelines
Advanced: Advanced students explore conditional formatting and custom number formats
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Data formatting exercise with before and after comparison
CT Skill Assessed: Visual presentation modifying data display for clarity and professional appearance

Dormitory — Retrieval (Paper)

Look at the menu in your hostel dining hall. Think about how you would format it as a spreadsheet with food items, prices, and availability using colors and borders.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher If you change a mark in one cell, should you manually recalculate the total in another cell? What if there were one hundred marks to update? There must be a way for the total to update automatically when you change any individual mark.

Students students realize that formulas with cell references automatically recalculate when source data changes. This saves enormous time.

Explore (10 min)

Teacher Enter the number ten in cell A1 and twenty in cell B1. In cell C1 type the formula equals A1 plus B1. Press Enter. Now change the value in A1 to fifteen. Watch C1 update automatically. Try other operations using multiply and divide symbols.

Students students discover that formulas recalculate instantly when referenced cells change. They experience the power of automatic calculation.

Explain (10 min)

Teacher Formulas start with equals sign and use arithmetic operators. Plus for addition, minus for subtraction, asterisk for multiplication, slash for division, and caret for exponent. Cell references like A1 refer to the value in that cell not the cell itself. Order of operations follows mathematics. Parentheses first, then exponents, then multiplication and division left to right, then addition and subtraction left to right. Always use cell references not fixed numbers so formulas update automatically.

Students students practice writing formulas with different operators. They learn to use parentheses to control calculation order.

Elaborate (10 min)

Teacher Create a simple bill calculator. Enter item names in column A, quantities in column B, and prices in column C. In column D write formulas to multiply quantity by price for each item. In a total cell write a formula to sum all line totals. Change a price and observe the total update.

Students students apply formulas to real-world calculation scenarios. They see how one change cascades through all dependent formulas.

Evaluate (5 min)

Teacher Write a formula to calculate the area of a rectangle using cell references for length and width. Write a formula to calculate average of five numbers in cells A1 through A5. Explain why cell references are better than fixed numbers in formulas.

Students students create practical formulas and articulate the benefit of dynamic recalculation over static values.

Cross-Curricular Connections

Mathematics: Calculate the total cost of school supplies for twenty students if each student needs five notebooks at fifteen rupees and three pens at ten rupees each.

Science: Explain how spreadsheet formulas follow the same order of operations as mathematics PEMDAS or BODMAS rules.
Language: Write a step-by-step tutorial explaining how to create a formula that calculates student average marks from five subject scores.

Digital Citizen Reminder: Spreadsheet formulas should be transparent and verifiable. Document your formulas so others can check your calculations. Do not use spreadsheets to present manipulated data as accurate.

Resources Needed:
- Spreadsheet software
- Formula reference card
- Practice data sheets
- Calculator for verification
- Textbook

Differentiation

Approaching: Provide formula templates with cell references already written where students identify the operation being performed
On Level: Students write standard formulas following operator and reference guidelines
Advanced: Advanced students explore absolute cell references with dollar signs for fixed values in formulas
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Formula writing exercise with calculation verification
CT Skill Assessed: Algorithmic thinking translating mathematical operations into spreadsheet formulas

Dormitory — Retrieval (Paper)

Write down the prices of five items you use every day. On paper calculate the total cost. Then think about how a spreadsheet formula would do the same calculation faster.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Writing a formula to add fifty student marks one cell at a time like equals A1 plus A2 plus A3 would be incredibly long. Is there a shorter way to add an entire column of numbers? Spreadsheet functions make this easy.

Students students imagine the impossibly long formula and appreciate that functions exist for common calculations like sum and average.

Explore (10 min)

Teacher Enter numbers from one to ten in cells A1 through A10. In cell A11 type equals SUM open parenthesis A1 colon A10 close parenthesis. The result should be fifty-five. Now try equals AVERAGE open parenthesis A1 colon A10 close parenthesis for the mean value.

Students students experience that functions handle entire ranges with simple syntax. They find SUM adds all values and AVERAGE calculates the arithmetic mean.

Explain (10 min)

Teacher Functions are pre-built formulas that perform common calculations. Syntax is function name followed by parentheses containing arguments. SUM adds all numbers in a range. AVERAGE calculates the mean. COUNT counts cells with numbers. MAX finds the highest value. MIN finds the lowest value. Ranges use colon to connect start and end cells like A1 colon A10 means all cells from A1 to A10. Functions can be combined in complex formulas.

Students students practice each function on sample data. They create a grade sheet using SUM for totals, AVERAGE for class average, and MAX for highest score.

Elaborate (10 min)

Teacher Create a complete marks sheet for five students in three subjects. Use SUM to calculate each student total. Use AVERAGE to find each student average. Use MAX and MIN to find highest and lowest marks in each subject. Add a class average row using AVERAGE across all students.

Students students apply multiple functions to create a comprehensive grade analysis. They understand how functions work together for complete data analysis.

Evaluate (5 min)

Teacher Write a SUM function to add cells B2 through B20. Write an AVERAGE function for the same range. Explain what COUNT would return for that range. Use MAX and MIN to find the range of values.

Students students demonstrate function syntax competence. They can apply appropriate functions for different analytical needs.

Cross-Curricular Connections

Mathematics: Calculate the average attendance percentage for a class if SUM of present days is two hundred and thirty out of two hundred and fifty total possible days.
Science: Explain how the AVERAGE function divides the sum of values by the count of values following the mathematical definition of arithmetic mean.
Language: Write a function reference guide for classmates explaining SUM, AVERAGE, COUNT, MAX, and MIN with example use cases.

Digital Citizen Reminder: Functions that process student data should be verified for accuracy. Always cross-check function results with manual calculations for critical data like grades and attendance.

Resources Needed:

- Spreadsheet software
- Function reference guide
- Practice data sets
- Grade sheet template
- Textbook

Differentiation

Approaching: Provide pre-entered data ranges where students practice writing SUM and AVERAGE functions
On Level: Students apply functions to standard data analysis exercises
Advanced: Advanced students explore nested functions like SUMIF and AVERAGEIF for conditional calculations
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Function writing quiz with practical data analysis task
CT Skill Assessed: Computational thinking using pre-built tools to solve mathematical problems efficiently

Dormitory — Retrieval (Paper)

Count how many hours you studied this week. Write the numbers for each day. Calculate the total hours using addition and the average hours per day using division.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Look at these two reports about student subject preferences. One shows a table of numbers. The other shows a colorful bar chart with the same data. Which one tells the story faster? What makes charts more powerful than tables for visual communication?

Students students immediately prefer the chart. They realize charts reveal patterns and comparisons that numbers alone hide.

Explore (10 min)

Teacher Enter subject names in column A and number of students in column B. Select the data and go to Insert menu. Choose Bar Chart. A chart appears. Now try Pie Chart and Line Graph with the same data. Notice how each chart type shows the data differently.

Students students discover that different chart types emphasize different aspects. Bar charts compare quantities. Pie charts show proportions. Line graphs show trends.

Explain (10 min)

Teacher Charts visualize data for better understanding. Bar charts compare quantities across categories using rectangular bars. Pie charts show proportions as slices of a circle where total is one hundred percent. Line graphs show trends over time using connected points. To create a chart select your data, choose the chart type, and customize labels and titles. Good charts have descriptive titles, labeled axes, clear legends, and appropriate colors that do not distract.

Students students practice creating each chart type. They learn to select data first then choose chart type that best represents their message.

Elaborate (10 min)

Teacher Create a chart showing the monthly rainfall in Karnataka from June to October. Use a bar chart for monthly comparison. Create a pie chart showing the percentage of rainfall in each month. Add a line graph showing the trend. Customize all three charts with proper titles and labels.

Students students create multiple chart types from the same data. They practice customizing chart elements for professional presentation.

Evaluate (5 min)

Teacher Choose the best chart type to show student marks distribution. Create a bar chart comparing five students total marks. Add a pie chart showing subject wise mark distribution. Explain why you chose each chart type.

Students students make informed chart selection decisions. They can justify their choices based on data characteristics and communication goals.

Cross-Curricular Connections

Mathematics: Calculate the percentage of total marks each subject represents for the pie chart using formula marks divided by total multiplied by one hundred.
Science: Explain how chart software converts spreadsheet data into visual elements by mapping values to bar heights, pie angles, and line positions.
Language: Write a data analysis report that uses charts to support conclusions about student performance trends.

Digital Citizen Reminder: Charts can misrepresent data by using misleading scales or truncated axes. Always start bar chart axes at zero and use consistent scales. Present data honestly in visual form.

Resources Needed:

- Spreadsheet software
- Sample data sets
- Chart type selection guide
- Color printing capability
- Textbook

Differentiation

Approaching: Provide pre-made data sets with suggested chart types for practice
On Level: Students create standard charts following data visualization guidelines
Advanced: Advanced students create combination charts and interactive charts with filters
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Chart creation exercise with type selection justification
CT Skill Assessed: Data visualization selecting appropriate visual representation for different data types

Dormitory — Retrieval (Paper)

Think about what you eat in the dining hall. Count how many times each food type appears in a week. Which chart would best show this data? Draw a simple version on paper.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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(..) => ..
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Engage (5 min)

Teacher Your teacher has a list of one hundred students with names, marks, and ranks. If asked to find the top ten students, how would you do it manually? Would sorting the list by marks make this task instant?

Students students realize that sorting transforms finding information from time-consuming search to instant visibility. They appreciate the power of ordered data.

Explore (10 min)

Teacher Enter a list of ten students with names and marks in a spreadsheet. Select the marks column. Go to Data menu and choose Sort Smallest to Largest. Observe how the entire rows rearrange keeping names matched with their marks. Now sort Largest to Smallest for ranking.

Students students discover that sorting rearranges entire rows to maintain data integrity. Each student's name stays with their correct mark.

Explain (10 min)

Teacher Sorting rearranges rows based on values in one or more columns. Ascending order puts smallest values first. Descending order puts largest values first. Sort by one column for simple ordering. Sort by multiple columns for complex organization like first by class then by marks within each class. Important rule is to include all columns when sorting so data stays aligned. Never sort only one column as it would separate data relationships.

Students students practice sorting by different columns and in different orders. They learn to always select the entire data range before sorting.

Elaborate (10 min)

Teacher Create a student database with columns for name, class, subject, and marks. Sort by marks to find top performers. Sort by class to group students. Sort by class first then by marks within each class for class-wise ranking. Identify the top three students in each class.

Students students practice multi-level sorting for complex data analysis. They discover that primary and secondary sort keys create meaningful groupings.

Evaluate (5 min)

Teacher Sort this data by student name alphabetically. Then sort by marks from highest to lowest. Create a class-wise sorted list showing top performer in each class. Explain what would happen if you sorted only the marks column without the names.

Students students demonstrate sorting competence and understand data integrity risks. They can predict problems from incorrect sorting.

Cross-Curricular Connections

Mathematics: Calculate how many comparisons a simple sort algorithm makes for ten items versus one hundred items.
Science: Explain how spreadsheet software uses comparison algorithms to determine the order of values during sorting.
Language: Write a sorting guide explaining when to use ascending versus descending order for different types of data analysis.

Digital Citizen Reminder: Sorting student data can reveal sensitive patterns. Be careful when sorting by attributes like gender, religion, or economic background. Ensure sorting serves educational purposes not discrimination.

Resources Needed:
- Spreadsheet software
- Practice data sets
- Sorting scenarios

- Data integrity checklist
- Textbook

Differentiation

Approaching: Provide small data sets with clear patterns for easy sorting practice
On Level: Students complete standard sorting exercises with single and multiple columns
Advanced: Advanced students explore custom sort orders and sorting by formula results
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Sorting exercise with data integrity verification
CT Skill Assessed: Data ordering organizing information to reveal patterns and enable efficient retrieval

Dormitory — Retrieval (Paper)

List five classmates with their heights. Sort the list from shortest to tallest. Who is the tallest? Who is in the middle?

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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(..) => ..
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Engage (5 min)

Teacher You have a spreadsheet with five hundred rows of student data from all KREIS schools. Your principal asks for only Class Seven students from Bidar district with marks above eighty. How would you find these records quickly without scrolling through everything?

Students students realize that filtering instantly shows only the records matching their criteria. This saves enormous time compared to manual searching.

Explore (10 min)

Teacher Enter a dataset with columns for name, class, district, and marks. Select the header row. Go to Data menu and enable AutoFilter. Click the dropdown arrow on the class column. Uncheck Select All and check only Seven. Notice only Class Seven rows appear. The row numbers turn blue indicating hidden rows.

Students students experience that filtering hides non-matching rows rather than deleting them. They see that filtered data is a temporary view of the full dataset.

Explain (10 min)

Teacher Filtering hides rows that do not match your criteria showing only relevant data. AutoFilter adds dropdown arrows to headers. Click the arrow to choose filter criteria. Text filters include contains, begins with, equals. Number filters include greater than, less than, between. Multiple column filters combine with AND logic meaning all conditions must be met. Clear Filter shows all rows again. Filters do not delete data they only hide it temporarily.

Students students practice applying filters to different columns. They learn to combine filters for complex queries like Class Seven AND marks greater than eighty.

Elaborate (10 min)

Teacher Create a school database with fifty students across three classes and two houses. Filter to show only Class Seven students. Then add a second filter for only Red House students. Now filter for students with marks above seventy-five. Clear each filter one at a time and observe the changes.

Students students practice layered filtering for complex data queries. They understand how combining filters narrows results progressively.

Evaluate (5 min)

Teacher Apply a filter to show only students from Bengaluru with marks between sixty and eighty. Explain what happens to the row numbers when a filter is active. How would you quickly see all data again after filtering?

Students students demonstrate filter competence. They understand visual indicators and can reset filters to see complete data.

Cross-Curricular Connections

Mathematics: Calculate the number of possible filter combinations when filtering three columns each with three options.
Science: Explain how filter algorithms scan through each row testing whether values match the specified criteria before deciding to show or hide that row.
Language: Write a data query guide for school administrators explaining how to filter student records for different reporting needs.

Digital Citizen Reminder: Filtered data views can be misleading if the audience does not realize rows are hidden. Always mention when presenting filtered data that it represents a subset of the full dataset.

Resources Needed:
- Spreadsheet software
- Large practice dataset
- Filter scenarios
- Data query examples
- Textbook

Differentiation

Approaching: Provide pre-formatted tables with AutoFilter already enabled for simple filter practice
On Level: Students apply standard filters following specific query requirements
Advanced: Advanced students explore advanced filter features and database query functions
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Filtering exercise with multi-criteria data query tasks
CT Skill Assessed: Data querying extracting specific information from large datasets based on conditions

Dormitory — Retrieval (Paper)

Think about your class. How many students are from your district? How many scored above fifty marks in the last test? These are filtering questions.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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(..) => ..
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Engage (5 min)

> **Teacher** Look at this marks sheet with five hundred numbers. It is impossible to quickly identify high and low performers. Now look at the same sheet with red for low marks, yellow for average, and green for high marks. How does color transform your ability to understand the data?

> **Students** students see that conditional formatting makes patterns visible instantly. They can spot high and low values without reading every number.

Explore (10 min)

> **Teacher** Enter student marks in a column. Select the marks range. Go to Format menu and choose Conditional Formatting. Apply a color scale from red through yellow to green. Notice how low numbers turn red, middle numbers turn yellow, and high numbers turn green. Try data bars to see length represent value.

> **Students** students experience that conditional formatting automatically assigns visual properties based on cell values. Colors update when data changes.

Explain (10 min)

Teacher Conditional formatting changes cell appearance based on their values. Color scales apply gradient colors from low to high values. Red-yellow-green is common for performance data. Data bars add colored bars proportional to cell value showing magnitude visually. Icon sets add symbols like arrows or traffic lights to indicate value ranges. Rules can use predefined scales or custom criteria like greater than seventy gets green. This makes patterns and outliers immediately visible.

Students students practice applying different conditional formatting types. They learn to choose appropriate visual indicators for their data.

Elaborate (10 min)

Teacher Create a class marks sheet and apply three conditional formatting rules. Use color scale for overall mark visualization. Use data bars for comparing student totals. Use icon sets to indicate pass or fail status. Test by changing values to see formatting update automatically.

Students students apply multiple conditional formatting rules to create informative data visualizations. They see how formatting reveals data stories.

Evaluate (5 min)

Teacher Apply a color scale to this temperature data from red for cold to blue for hot. Create data bars for these sales figures. Add traffic light icons where green is above target, yellow is at target, and red is below target.

Students students apply conditional formatting appropriate to data context. They demonstrate understanding of when each format type is most effective.

Cross-Curricular Connections

Mathematics: Calculate the color value assignments for a three-color scale if the minimum value is twenty and maximum is one hundred.
Science: Explain how conditional formatting algorithms interpolate colors across a value range to assign appropriate hue and brightness to each cell.
Language: Write a data visualization guide explaining when to use color scales versus data bars versus icon sets for different data types.

Digital Citizen Reminder: Conditional formatting colors can create false urgency or calm. Red does not always mean bad and green does not always mean good. Choose color schemes that match the actual meaning of your data.

Resources Needed:
- Spreadsheet software
- Sample data sets
- Conditional formatting reference
- Color scale examples
- Textbook

Differentiation

Approaching: Provide pre-entered data with step-by-step instructions for applying specific conditional formatting rules
On Level: Students apply standard conditional formatting following data visualization guidelines
Advanced: Advanced students create custom conditional formatting rules using formulas for complex conditions
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Conditional formatting application exercise with visual analysis
CT Skill Assessed: Visual analytics using automatic visual properties to reveal data patterns and anomalies

Dormitory — Retrieval (Paper)

Look at a traffic light tonight. What do the three colors mean? In data, what would green, yellow, and red indicate for student attendance?

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Your teacher enters student marks but accidentally types one hundred and fifty instead of fifty. The average calculation becomes wrong. How could the spreadsheet prevent this error at the moment of data entry rather than finding it later?

Students students realize that data validation prevents errors before they happen. This is more efficient than correcting mistakes afterward.

Explore (10 min)

Teacher Select a column where marks will be entered. Go to Data menu and choose Data Validation. Set the criteria to whole number between zero and one hundred. Try entering one hundred and fifty. Notice an error message appears preventing the invalid entry. Now try entering eighty and it accepts it.

Students students experience that data validation rejects invalid entries immediately. They see how validation rules enforce data quality at the source.

Explain (10 min)

Teacher Data validation controls what users can enter in cells. Settings include number ranges like zero to one hundred for marks, date ranges for valid date periods, text length limits, and custom formulas for complex rules. Dropdown lists provide predefined options ensuring consistent spelling and terminology. Input messages appear when a cell is selected guiding users. Error alerts appear when invalid data is entered with options to retry or cancel. This prevents data quality problems before they occur.

Students students practice setting different validation rules. They create dropdown lists for class selection and number ranges for marks.

Elaborate (10 min)

Teacher Create a student data entry form with validation. Column A for names with text length maximum fifty. Column B for class with dropdown list of classes. Column C for marks with number between zero and one hundred. Column D for grade with dropdown list. Test each validation by trying invalid entries.

Students students build complete validated data entry systems. They understand that validation creates reliable data for analysis.

Evaluate (5 min)

Teacher Set up validation to accept only dates in the current school year. Create a dropdown list with five subject names. Explain what happens when a user tries to enter invalid data. How would you help a user who sees an error message?

Students students demonstrate validation setup competence. They can guide users through validation error resolution.

Cross-Curricular Connections

Mathematics: Calculate how many different validation rule combinations are possible with number range, date range, and text length options.
Science: Explain how dropdown list validation uses predefined arrays of valid options to check user input against allowed values.
Language: Write a user guide explaining data validation rules for a school data entry form so clerks understand the requirements.

Digital Citizen Reminder: Data validation improves data integrity which supports accurate decision making. Poor data leads to poor decisions. Always validate data at the point of entry for reliable school management.

Resources Needed:

- Spreadsheet software
- Validation examples
- Dropdown list templates
- Error message examples
- Textbook

Differentiation

Approaching: Provide templates with pre-set validation rules where students test and document the behavior
On Level: Students create standard validation rules following data quality guidelines
Advanced: Advanced students create dependent dropdown lists where options change based on previous selections
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Data validation setup and testing exercise
CT Skill Assessed: Input validation ensuring data quality through rule-based entry constraints

Dormitory — Retrieval (Paper)

Think about forms you fill out online. Have you ever been prevented from entering wrong data? What validation rule was probably applied?

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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(..) => ..
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Engage (5 min)

Teacher You have a spreadsheet with one thousand rows of sales data from all KREIS school stores. The principal wants to know total sales per school and average sale per item. Creating formulas for each school would take days. Is there a tool that can summarize large data instantly?

Students students realize that pivot tables can summarize thousands of rows into useful summaries in seconds. They want to learn this powerful tool.

Explore (10 min)

Teacher Open a sample sales dataset with columns for school, item, quantity, and price. Select all data. Go to Insert menu and choose Pivot Table. A dialog appears asking where to place it. Choose a new worksheet. Now drag school to Rows area and price to Values area. Watch the summary appear instantly.

Students students experience that pivot tables automatically group and summarize data. They see that dragging fields to different areas changes the summary view.

Explain (10 min)

Teacher Pivot tables summarize large datasets into meaningful reports. They have four areas. Rows define what to group by like school names. Columns define what to subdivide by like item types. Values define what to calculate like sum of price. Filters define what subset to include. You can drag fields between areas to change the summary. Double-click values to see the detail behind any summary number. Pivot tables update when source data changes.

Students students practice creating pivot tables with different field arrangements. They learn to read and interpret pivot table summaries.

Elaborate (10 min)

Teacher Create a pivot table from student marks data. Put class in Rows and subject in Columns and marks in Values using average function. This creates a class-wise subject-wise average marks table. Add a filter for house to see averages for specific houses only.

Students students create multi-dimensional summaries using pivot tables. They discover how quickly pivot tables answer complex data questions.

Evaluate (5 min)

Teacher Create a pivot table showing total marks per student. Create another showing average marks per subject. Explain what each summary tells the principal that raw data does not.

Students students demonstrate pivot table creation and interpretation. They can explain the analytical value of summarized data.

Cross-Curricular Connections

Mathematics: Calculate how many formula cells would be needed to create a class-wise subject-wise average table manually versus using one pivot table.
Science: Explain how pivot table algorithms group rows with matching values and apply aggregate functions to calculate summaries.
Language: Write a report comparing raw data presentation versus pivot table summaries for a principal making school improvement decisions.

Digital Citizen Reminder: Pivot tables reveal patterns in student data that should be handled sensitively. Avoid sharing pivot table analyses that could stigmatize specific groups of students based on demographic data.

Resources Needed:
- Spreadsheet software
- Large sample dataset
- Pivot table field list guide
- Summary examples
- Textbook

Differentiation

Approaching: Provide pre-built pivot tables where students practice reading and interpreting the summaries
On Level: Students create basic pivot tables following standard field placement guidelines
Advanced: Advanced students explore calculated fields and pivot charts for dynamic reporting
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Pivot table creation and data interpretation exercise
CT Skill Assessed: Data aggregation summarizing large datasets into meaningful actionable insights

Dormitory — Retrieval (Paper)

Count how many hours you spent on each subject this week. How would you summarize this data to show which subject you studied most?

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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`(..) => ..`

Engage (5 min)

> **Teacher** Your school wants to improve student performance. You have marks data from the last three exams. How would you analyze this data to find which subjects need more attention, which students are improving, and which are falling behind?

> **Students** students realize that raw data needs analysis to become actionable insights. They plan to use all the spreadsheet skills they have learned.

Explore (10 min)

> **Teacher** Open a comprehensive student marks dataset. Apply sorting to rank students. Use filters to focus on specific classes. Calculate averages with functions. Create charts to visualize trends. Apply conditional formatting to highlight concerns. Notice how each tool reveals different insights.

> **Students** students experience that combining multiple analysis tools provides complete understanding. Sorting shows rankings, filtering isolates groups, functions calculate metrics, and charts reveal patterns.

Explain (10 min)

> **Teacher** Data analysis combines multiple techniques. First organize data with proper formatting and validation. Then sort to see rankings. Filter to focus on specific groups. Calculate key metrics using SUM, AVERAGE, MAX, MIN. Visualize patterns with appropriate charts. Apply conditional formatting to highlight important values. Finally interpret findings and write conclusions. The goal is to transform raw data into actionable insights for decision making.

> **Students** students follow a structured analysis workflow on sample data. They learn that systematic analysis produces reliable insights.

Elaborate (10 min)

> **Teacher** Define three-exam marks data for your class. Sort students by improvement between exam one and exam three. Create charts showing subject-wise class averages. Identify the three students with most improvement and three with most decline. Write a one-page analysis report with recommendations.

Students students complete comprehensive analysis projects. They practice data interpretation and report writing for real-world application.

Evaluate (5 min)

Teacher Present your data analysis findings to the class. Explain what tools you used and why. Describe three key findings from your analysis. Recommend two actions the school could take based on your findings.

Students students present data-driven recommendations. They demonstrate ability to translate data patterns into practical suggestions.

Cross-Curricular Connections

Mathematics: Calculate improvement percentage between exam scores using formula new score minus old score divided by old score multiplied by one hundred.
Science: Explain how statistical analysis of educational data helps identify learning patterns and inform teaching strategies.
Language: Write an executive summary of your data analysis for the school principal highlighting key findings and recommendations.

Digital Citizen Reminder: Data analysis should lead to helping students not labeling them. Present findings constructively focusing on support needed rather than deficits identified. Protect student privacy in all reports.

Resources Needed:

- Comprehensive student dataset
- Analysis template
- Chart examples
- Report writing guide
- Textbook

Differentiation

Approaching: Provide structured analysis worksheets with guiding questions for each analysis step
On Level: Students complete standard analysis following the organized workflow
Advanced: Advanced students explore statistical functions like standard deviation for deeper analysis
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Data analysis report with charts and written findings
CT Skill Assessed: Analytical reasoning combining multiple data processing techniques to extract insights

Dormitory — Retrieval (Paper)

Think about your marks in the last three tests. Which subject improved the most? Which declined? Write two possible reasons for each trend.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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(..) => ..
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Engage (5 min)

Teacher Imagine you receive five thousand rupees pocket money each month from your family at KREIS school. You need to manage food, supplies, entertainment, and savings. How would you track every rupee and ensure you do not run out of money before the month ends?

Students students realize that budgeting requires careful tracking and planning. They want to create a system that helps them manage money wisely.

Explore (10 min)

Teacher Open a blank spreadsheet. Create columns for Date, Description, Category, Amount In, and Amount Out. Enter some sample income and expense entries. In a summary area use SUM to calculate total income and total expenses. Calculate the balance as income minus expenses.

Students students experience creating a basic budget structure. They see how formulas automatically calculate running totals.

Explain (10 min)

Teacher A budget tracks income and expenses over a period. Structure includes income section listing all money received, expense section listing all money spent, category column for organizing expenses like food, transport, supplies, entertainment, savings, and summary section with totals and balance. Formulas needed include SUM for totals, subtraction for balance, percentage formulas for category spending as percent of total. Charts help visualize where money goes. Update the budget daily for accuracy.

Students students build complete budget spreadsheets. They practice entering realistic KREIS school expenses and calculating meaningful summaries.

Elaborate (10 min)

Teacher Create a monthly budget for managing five thousand rupees at KREIS school. Categorize expenses into food, stationery, transport, entertainment, and savings. Enter daily transactions for one simulated week. Use formulas to calculate category totals and percentage of total spending. Create a pie chart showing expense distribution. Write recommendations for reducing unnecessary spending.

Students students create functional budget tools. They practice realistic financial planning and data visualization for personal finance management.

Evaluate (5 min)

Teacher Present your budget spreadsheet. Show the total income, total expenses, and remaining balance. Create a chart showing spending by category. Explain two ways the budget could be improved for next month.

Students students demonstrate budget creation and financial analysis skills. They can identify spending patterns and suggest improvements.

Cross-Curricular Connections

Mathematics: Calculate the percentage of total spending that goes to food if food expenses are one thousand two hundred rupees out of four thousand five hundred total expenses.
Science: Explain how budget tracking relates to economic principles of scarcity, opportunity cost, and resource allocation.
Language: Write a financial literacy guide for classmates explaining how to create and maintain a personal monthly budget.

Digital Citizen Reminder: Financial data is extremely sensitive. Protect your budget files with passwords. Never share your financial information with strangers online. Be cautious with financial apps and websites.

Resources Needed:
- Spreadsheet software
- Budget template
- KREIS expense examples
- Chart tools
- Financial literacy guide
- Textbook

Differentiation

Approaching: Provide budget templates with categories and formulas already set up where students enter transaction data
On Level: Students create complete budget spreadsheets from scratch following guidelines
Advanced: Advanced students create annual budgets with monthly tracking and trend analysis
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Budget project rubric with formula accuracy, chart quality, and analysis depth
CT Skill Assessed: Financial planning creating systems to track and analyze monetary resources

Dormitory — Retrieval (Paper)

Tonight write down everything you spent money on today. Estimate how much you will spend this month. Think about what categories these expenses belong to.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Chapter 5 — Programming with Scratch

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Engage (5 min)

Teacher Think back to when you first used Scratch. What were the first three blocks you ever used? Can you remember what they did? Today we will refresh our memory and prepare for more advanced programming.

Students Students recall green flag event, move steps block, and say hello block. They feel nostalgic and ready to build on their foundation.

Explore (10 min)

Teacher Open Scratch without any instructions. Try to recreate your very first project from last year. Use whatever blocks you remember. If you get stuck, look at the block palette and see if any blocks look familiar.

Students Students struggle with some forgotten concepts but recover quickly as they see familiar blocks. They realize they remember more than they thought.

Explain (10 min)

Teacher Scratch has a stage where the program runs, sprites that perform actions, and scripts that control sprite behavior. Block categories include Events for starting scripts, Motion for movement, Looks for appearance, Sound for audio, Control for loops and conditions, Sensing for interaction, Operators for math and logic, and Variables for storing data. Each block snaps together like puzzle pieces to build programs.

Students Students create a reference sheet listing all block categories with two example blocks from each. They rebuild a simple animation from memory.

Elaborate (10 min)

Teacher Recreate a project that makes a character walk across the screen, say hello, and change color. Use at least five different block categories. Add a second sprite with its own script. Make both sprites respond to the green flag click.

Students Students rebuild projects using accumulated knowledge. They practice combining blocks from different categories for richer interactions.

Evaluate (5 min)

Teacher List all eight block categories with one block from each. Explain what the green flag does. Describe the difference between a forever loop and a repeat loop. Debug this broken script and explain what was wrong.

Students Students demonstrate fundamental Scratch knowledge. They can identify, explain, and fix basic program elements.

Cross-Curricular Connections

Mathematics: Calculate the total number of steps a sprite moves if it executes move ten steps inside a repeat twenty loop.
Science: Explain how the computer processes Scratch blocks from top to bottom within each script following sequential execution.
Language: Write a program description explaining step by step what each block does in a simple animation script.

Digital Citizen Reminder: Scratch projects are shared publicly by default. Keep your real name and personal information private when sharing projects online.

Resources Needed:
- Scratch software or website
- Previous year projects
- Block reference guide
- Debugging examples
- Textbook

Differentiation

Approaching: Provide block matching cards where students connect block pictures to their functions
On Level: Students complete standard review exercises rebuilding basic projects
Advanced: Advanced students review advanced blocks like clones and custom blocks for upcoming lessons
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Block identification quiz and project rebuild test
CT Skill Assessed: Knowledge retrieval recalling previously learned concepts for application

Dormitory — Retrieval (Paper)

Open Scratch on any available device tonight and try to make a cat sprite say your name. Remember how to attach blocks together.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Imagine a game where your score increases every time you catch a coin. Where does the game store the current score? How does it remember the score even after the coin disappears? What programming tool handles this memory?

Students Students realize that variables act as memory storage for programs. They understand that games need to remember scores, lives, and levels.

Explore (10 min)

Teacher Open Scratch and create a new variable called score. Notice it appears in the variables palette and optionally on the stage. Use set score to zero block. Then use change score by one block. Click these blocks and watch the score variable update on the stage.

Students Students experience that variables hold values that can be changed. They see the variable value update in real-time on the stage.

Explain (10 min)

Teacher Variables are named containers that store data values. Create a variable with the Make a Variable button. Set block assigns an initial value. Change block adds to the current value. Variables can hold numbers for scores and levels, text for names and messages, or true or false for game states. For all sprites scope means every sprite can read and change the variable. For this sprite only scope means only that sprite can access the variable.

Students Students create variables for different purposes. They practice using variables in calculations and understand scope differences.

Elaborate (10 min)

Teacher Create a counter program. Make a variable called count. Set it to zero. Create two buttons using sprites. One button increases count by one when clicked. The other decreases count by one when clicked. Display the count value on the stage. Add a reset button that sets count back to zero.

Students Students build practical variable applications. They learn that variables enable user interaction and dynamic program state.

Evaluate (5 min)

Teacher Create a variable called timer that counts down from sixty. Use the change and set blocks correctly. Explain when you would use for all sprites versus for this sprite only scope.

Students Students demonstrate variable creation and scope understanding. They apply variables to practical timing scenarios.

Cross-Curricular Connections

Mathematics: Calculate the final value of a variable if it starts at zero and is changed by three, then changed by negative five, then changed by eight.
Science: Explain how computer memory stores variable values using addresses and how Scratch manages variable allocation automatically.

Language: Write a flowchart showing how a score variable changes through set, increase, and decrease operations in a game.

Digital Citizen Reminder: Variables that store personal information like names and scores should be handled carefully. Do not create variables that collect personal data from other users without their knowledge.

Resources Needed:
- Scratch software
- Variable reference sheet
- Counter activity guide
- Scope diagram
- Textbook

Differentiation

Approaching: Provide variable creation exercises with step-by-step instructions and visual block guides
On Level: Students create standard variable applications following programming guidelines
Advanced: Advanced students explore lists as collections of variables and create scoreboards with multiple stored values
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Variable creation and usage practical exercise
CT Skill Assessed: State management using variables to track and modify program data

Dormitory — Retrieval (Paper)

Write down three things a video game needs to remember while playing. Think about score, lives, and level number. These would all be variables.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

—

```
(..) => ..
```

Engage (5 min)

Teacher Think about getting dressed in the morning. If it is raining you take an umbrella. If it is sunny you wear sunglasses. Your brain makes decisions based on conditions. How do we teach a computer to make similar decisions?

Students Students relate programming conditions to daily decisions. They realize that computers need explicit rules to make choices.

Explore (10 min)

Teacher Open Scratch and create a simple program. Make the sprite move ten steps. Now wrap it in an if-then block with the condition touching edge. Inside the if put a turn fifteen degrees block. Run it and watch what happens when the sprite reaches the edge.

Students Students see that the sprite only turns when the condition is true. They experience conditional execution firsthand.

Explain (10 min)

Teacher If-then blocks check a condition and execute the enclosed blocks only if the condition is true. If-then-else blocks provide two paths. If true the first set of blocks runs. If false the second set runs. Comparison operators include equals, greater than, less than. Logical operators include and for both conditions true, or for either condition true, and not for reversing a condition. Conditions can check sprite positions, variable values, key presses, and other states.

Students Students practice creating programs with decisions. They learn to choose appropriate conditions and operators for different scenarios.

Elaborate (10 min)

Teacher Create a quiz program. Ask a question using ask block. Use if-then-else to check if the answer equals the correct response. If correct increase score by one and say correct for two seconds. If wrong say try again and decrease lives by one.

Students Students build interactive programs with conditional logic. They experience how conditions create branching program behavior.

Evaluate (5 min)

Teacher Write a program that makes a sprite grow big if the space key is pressed and shrink if the down arrow is pressed. Explain the difference between if-then and if-then-else blocks. Create a condition that checks if score is greater than ten.

Students Students demonstrate condition competence. They understand when each block type is appropriate.

Cross-Curricular Connections

Mathematics: Calculate how many different paths a program can take with three if-else blocks each checking different conditions.
Science: Explain how comparison operators work at the binary level by comparing bits representing numbers in computer memory.
Language: Write a decision tree diagram showing all possible paths through a program with three conditional checks.

Digital Citizen Reminder: Conditional logic in programs can be biased if the conditions are based on stereotypes. Ensure that game and program conditions treat all users fairly regardless of background.

Resources Needed:
- Scratch software
- Condition examples
- Flowchart templates
- Comparison operator reference
- Textbook

Differentiation

Approaching: Provide pre-written conditions where students identify whether each is true or false for given values
On Level: Students create standard if-then and if-then-else programs
Advanced: Advanced students explore nested conditions and compound logical operators for complex decision making
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Condition creation exercise with program testing
CT Skill Assessed: Decision logic implementing conditional branching based on evaluated criteria

Dormitory — Retrieval (Paper)

Write three if-then sentences about your morning routine. Example if it is morning then brush teeth. Think about what conditions you check every day.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

—

`(..) => ..`

Engage (5 min)

Teacher Think about ordering food at a restaurant. First you decide if you want vegetarian or non-vegetarian. Then within that choice you decide if you want spicy or mild. You make a decision inside another decision. How would you program this layered decision making?

Students Students see that real decisions often happen in layers. They realize nested conditions model complex real-world choices.

Explore (10 min)

Teacher Open Scratch and create a program with an if-then block that checks if the mouse is down. Inside that if put another if-then that checks if the mouse is touching a specific sprite. Notice that the inner condition only runs when the outer condition is already true.

Students Students experience that nested conditions create gates. The inner condition cannot be checked unless the outer condition passes first.

Explain (10 min)

Teacher Nested conditions place one if block inside another. The outer condition is checked first. If it is false the inner condition is never checked. If it is true the inner condition is then evaluated. This creates a hierarchy of decisions. The program flows from outer to inner conditions. Each level narrows the requirements. Nested conditions can be three or four levels deep but too many levels make code hard to read. Alternatives include combining conditions with logical operators.

Students Students practice building nested condition programs. They learn to choose between nesting and combining conditions based on clarity.

Elaborate (10 min)

Teacher Create a character customization program. First check if a costume button is clicked. Inside that check which character type is selected. Inside that check which color is chosen. Apply the complete selection to the sprite. Make sure each nesting level has a clear purpose.

Students Students build multi-level decision programs. They practice organizing complex logic in readable nested structures.

Evaluate (5 min)

Teacher Trace through this nested condition program. What happens when the outer condition is true and the inner is false? Create a program with three nesting levels. Explain why nesting was better than combining all conditions in one if block for your example.

Students Students can trace program execution through nested conditions. They justify their design choices with reasoning.

Cross-Curricular Connections

Mathematics: Calculate the maximum number of outcomes possible with four nested if-else blocks each having two branches.
Science: Explain how nested conditional logic relates to hierarchical classification systems in biology where organisms are classified from kingdom to species.
Language: Write a debugging guide explaining common mistakes when tracing nested condition execution order.

Digital Citizen Reminder: Nested conditions in algorithms can create hidden biases if the nesting order prioritizes certain groups. Consider how algorithmic decisions affect different users equitably.

Resources Needed:

- Scratch software
- Nested condition examples
- Decision tree templates
- Tracing worksheets
- Textbook

Differentiation

Approaching: Provide partially completed nested condition programs where students fill in the missing inner blocks
On Level: Students create standard nested condition programs following guidelines
Advanced: Advanced students explore deeply nested conditions and refactor them into cleaner logical alternatives
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Nested condition program creation and tracing exercise
CT Skill Assessed: Hierarchical decision making implementing multi-level conditional logic

Dormitory — Retrieval (Paper)

Think about choosing what to eat. First decide sweet or savory. Then within that decide specific item. Write this as two nested if-then statements.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

—

```
(..) => ..
```

Engage (5 min)

Teacher Think about washing dishes. You keep washing until all dishes are clean. You do not count how many dishes there are. You stop when a condition is met. How would you program a computer to repeat until a condition changes rather than repeating a fixed number of times?

Students Students realize that repeat-until loops are more flexible than repeat loops. They understand condition-based repetition matches many real activities.

Explore (10 min)

Teacher Open Scratch and create a repeat-until loop that makes a sprite move until it touches the edge. Use the touching edge condition. Add a turn block inside the loop. Watch the sprite move and turn until it reaches the boundary.

Students Students see that repeat-until keeps executing while the condition is false and stops when it becomes true. They observe the sprite behavior change dynamically.

Explain (10 min)

Teacher Repeat-until loops execute blocks repeatedly until a condition becomes true. The condition is checked before each iteration. When the condition is true the loop stops. This differs from repeat which runs a fixed number of times. Repeat-until is better when you do not know how many iterations are needed. Important safety rule always ensure the condition will eventually become true to avoid infinite loops. Infinite loops freeze the program and require force quitting.

Students Students practice creating repeat-until loops with various conditions. They learn to verify that their termination conditions will eventually be met.

Elaborate (10 min)

Teacher Create a maze game. The sprite starts at the beginning and the player uses arrow keys to navigate. Use a repeat-until loop that runs until the sprite touches the goal. Inside the loop check key presses and move accordingly. Add a win message when the goal is reached.

Students Students build interactive games using repeat-until loops. They experience how loops create continuous gameplay that ends at a specific condition.

Evaluate (5 min)

Teacher Create a repeat-until loop that counts down from sixty to zero. Explain why this loop will not run forever. Create a loop that stops when two conditions are both true. Identify the termination condition in each program.

Students Students demonstrate understanding of loop termination. They can verify that their loops will end correctly.

Cross-Curricular Connections

Mathematics: Calculate how many iterations a repeat-until loop makes if it starts at one hundred and decreases by seven each iteration until the value is less than zero.
Science: Explain how the repeat-until loop compares to a while loop in other programming languages and why condition order matters.
Language: Write a real-world algorithm using repeat-until logic for making tea until it tastes right or studying until understanding is achieved.

Digital Citizen Reminder: Infinite loops can consume computer resources and slow down networks. Always test loops carefully to ensure they terminate. Do not create programs that intentionally freeze others devices.

Resources Needed:
- Scratch software
- Loop examples

- Infinite loop warning
- Termination condition guide
- Textbook

Differentiation

Approaching: Provide loop completion exercises where students predict how many iterations will occur before termination
On Level: Students create standard repeat-until programs with verified termination conditions
Advanced: Advanced students combine repeat-until with nested conditions for complex interactive behaviors
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Loop creation and termination verification exercise
CT Skill Assessed: Iterative processing implementing repetition with clear stopping criteria

Dormitory — Retrieval (Paper)

Think about a task you repeat every day until it is done. Write a repeat-until statement describing it. What is the condition that tells you to stop?

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

—

```
(..) => ..
```

Engage (5 min)

Teacher Think about your class attendance register. It stores fifty student names in a specific order. You can add new names, remove names, and look up any name by position. How would you create this kind of ordered collection in a program?

Students Students realize that variables store single values but lists store many values. They see lists as ordered collections like registers and inventories.

Explore (10 min)

Teacher Open Scratch and create a new list called names. Add your name using add thing to names block. Add three classmates names. Use delete block to remove the second item. Use insert block to add a name at position two. Look at how the list changes on the stage.

Students Students experience list manipulation operations. They see that lists display on the stage and can be modified dynamically.

Explain (10 min)

Teacher Lists store ordered collections of values. Create a list with Make a List button. Add block appends an item to the end. Delete block removes an item by position. Insert block adds an item at a specific position. Replace block changes an item at a position. Item of block reads a value by position. Length of block counts items. Contents of block reports all items. Lists use positions starting from one not zero. Loops can process every item in a list using the length to determine iterations.

Students Students practice all list operations. They learn to read, write, and modify list contents programmatically.

Elaborate (10 min)

Teacher Create a to-do list application. Make a list called tasks. Add buttons to add new tasks, mark tasks complete by deleting them, and display all tasks. Use a loop to number each task when displaying. Count the remaining tasks using length.

Students Students build practical list-based applications. They experience how lists enable dynamic data management in programs.

Evaluate (5 min)

Teacher Create a list of five favorite foods. Delete the third item. Insert a new item at position one. Replace the last item. Write a loop that says each item. Explain what happens if you try to access item ten in a list with only five items.

Students Students demonstrate list manipulation competence. They understand list boundaries and error handling.

Cross-Curricular Connections

Mathematics: Calculate the number of different ways to arrange five items in a list using mathematical permutations.
Science: Explain how computer memory stores list elements in sequential addresses and how the program accesses them by position index.
Language: Write a comparison of lists versus variables explaining when to use each storage type with three example scenarios.

Digital Citizen Reminder: Lists containing personal information like names and addresses are sensitive data. Protect list data and never share personal lists publicly without consent.

Resources Needed:
- Scratch software
- List operations reference
- To-do list activity guide
- Array concept diagram
- Textbook

Differentiation

Approaching: Provide pre-made lists where students practice reading and modifying with guided operations
On Level: Students create and manipulate lists following standard operations
Advanced: Advanced students explore parallel lists and create databases with multiple related lists
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: List manipulation exercise with practical application task
CT Skill Assessed: Collection management organizing and processing ordered groups of data

Dormitory — Retrieval (Paper)

Write a list of five things you need to pack when going home for holidays. Number each item. This is how a list works in programming.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

—

```
(..) => ..
```

Engage (5 min)

Teacher Think about a game where enemies appear from the right side of the screen repeatedly. Do you create a hundred separate sprites or do you create one enemy sprite that copies itself? How would a computer handle making multiple copies of the same character?

Students Students realize that creating hundreds of sprites would be inefficient. They wonder how games create many instances of the same enemy efficiently.

Explore (10 min)

Teacher Open Scratch and create a coin sprite. Add a green flag event and a forever loop. Inside the loop add create clone of myself block and wait one second. Now add when I start as clone block with a go to random position and fall down script. Watch clones appear and fall.

Students Students experience that clones behave like copies of the original sprite. They see that each clone runs its own when I start as clone script.

Explain (10 min)

Teacher Clones are temporary copies of a sprite created during program execution. Use create clone of block to make a new clone. The clone starts running scripts under when I start as clone hat block. Each clone is independent and can have different position and costume. Clones can create more clones but Scratch limits total clones to three hundred. Use delete this clone block to remove clones when they are no longer needed. Clones are destroyed when the program stops.

Students Students practice creating and managing clones. They learn to control clone creation rate and deletion for efficient programs.

Elaborate (10 min)

Teacher Create a rain simulation. Make a raindrop sprite that clones itself repeatedly. Each clone starts at the top of the screen in a random horizontal position and falls to the bottom. When a clone reaches the bottom it is deleted. Add slight wind effect by moving clones sideways as they fall.

Students Students build particle simulation effects using clones. They experience how clones create complex visual effects efficiently.

Evaluate (5 min)

Teacher Create a bullet system where a spaceship sprite clones a bullet when space is pressed. Each bullet moves upward and is deleted when it reaches the top. Explain why clones are better than creating individual bullet sprites. How would you limit the number of clones on screen?

Students Students demonstrate clone management competence. They understand memory implications and practical limitations.

Cross-Curricular Connections

Mathematics: Calculate the maximum number of clones a rain simulation could have if it creates one clone per second and clones last for five seconds.
Science: Explain how object-oriented programming uses classes and instances which are similar to Scratch sprites and clones.
Language: Write a technical specification for a game enemy system describing clone creation, behavior, and cleanup requirements.

Digital Citizen Reminder: Clones can be used to create spam or flood systems with unwanted elements. Responsible programming means managing clone counts to avoid overwhelming computer resources.

Resources Needed:
- Scratch software
- Clone examples
- Rain simulation template
- Object concept diagram
- Textbook

Differentiation

Approaching: Provide clone creation exercises with visual guides showing sprite to clone relationships
On Level: Students create standard clone applications following guidelines
Advanced: Advanced students explore clone communication using broadcasts for coordinated clone behavior
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Clone creation and management practical exercise
CT Skill Assessed: Object instantiation creating and managing multiple instances of a template

Dormitory — Retrieval (Paper)

Think about a game you play. How many similar characters appear at the same time? Are they individual sprites or copies of the same sprite?

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

—

```
(..) => ..
```

Engage (5 min)

Teacher Watch this animation where a sprite draws a perfect square as it moves. Now it draws a spiral. Now a flower pattern. The sprite is creating art using only movement commands. How can moving and turning create beautiful drawings?

Students Students are amazed that simple movement can create complex art. They want to learn the drawing commands.

Explore (10 min)

Teacher Open Scratch and add the pen extension from the extensions button. Create a green flag event. Add erase all to clear previous drawings. Add pen down to start drawing. Make the sprite move one hundred steps and turn ninety degrees four times. A square appears.

Students Students discover that pen down activates drawing as the sprite moves. They see that loops create geometric shapes automatically.

Explain (10 min)

Teacher The pen extension lets sprites draw on the stage. Pen down starts drawing a trail as the sprite moves. Pen up stops drawing. Set pen color changes the drawing color. Set pen size changes line thickness. Erase all clears the stage. Change pen color by number creates rainbow effects. Combine pen blocks with motion and loops to create geometric art. Repeat loops with turn angles create polygons and flowers. Nested loops create complex recursive patterns.

Students Students practice creating geometric shapes with pen commands. They learn to calculate turn angles for different polygons like three hundred sixty divided by number of sides.

Elaborate (10 min)

Teacher Create a spirograph art generator. Use nested loops where the outer loop turns a small angle and the inner loop draws a circle. Change pen color in each iteration for rainbow effect. Experiment with different angle combinations to discover unique patterns.

Students Students create complex generative art using nested loops and pen commands. They discover mathematical beauty in programming.

Evaluate (5 min)

Teacher Draw a regular hexagon using pen and motion blocks. Create a pattern with eight triangles rotated around a center point. Explain why the turn angle must be exactly three hundred sixty divided by the number of sides for regular polygons.

Students Students demonstrate geometric drawing competence. They connect programming to mathematical concepts.

Cross-Curricular Connections

Mathematics: Calculate the exterior angle for a regular polygon with seven sides. How many times must the sprite turn to complete the polygon?
Science: Explain how turtle graphics originated at MIT and how pen drawing relates to vector graphics using mathematical paths.
Language: Write a guide explaining how to create five different geometric shapes using pen blocks with specific angle calculations.

Digital Citizen Reminder: Pen art created in Scratch is original creative work. Share your art with attribution. Respect others creative coding art and do not copy without credit.

Resources Needed:
- Scratch software with pen extension
- Geometry reference
- Pattern examples
- Angle calculation guide
- Textbook

Differentiation

Approaching: Provide pre-written pen programs where students modify parameters to change the output
On Level: Students create standard geometric shapes following mathematical guidelines
Advanced: Advanced students explore fractal patterns and recursive drawing algorithms
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Geometric art creation with mathematical explanation
CT Skill Assessed: Computational art using algorithms to create visual patterns and designs

Dormitory — Retrieval (Paper)

Draw a simple geometric pattern on paper using a ruler and protractor. Think about how you would tell a computer to draw the same pattern using only move and turn commands.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

—

`(..) => ..`

Engage (5 min)

Teacher Think about your favorite song. It is made of individual notes played in a specific rhythm. Can you program a computer to play a melody you know? What blocks would you need to control pitch and timing?

Students Students want to recreate familiar melodies. They realize that music is a sequence of notes with specific durations.

Explore (10 min)

Teacher Open Scratch and add the music extension. Play a middle C note for one beat. Then play D for one beat. Then E. Then F. Then G. You just played the first five notes of a scale. Try changing note values and durations to create different melodies.

Students Students discover that music blocks control pitch with note numbers and duration with beats. They experiment with creating simple melodies.

Explain (10 min)

Teacher The music extension provides blocks for creating audio. Play note block specifies pitch using note numbers where sixty is middle C. Play note for beats controls duration where one beat is one quarter note at the current tempo. Set tempo controls speed in beats per minute. Drum blocks add percussion sounds. Rest for beats creates silence between notes. Combine music blocks with say blocks to display lyrics while music plays. Loops can repeat musical phrases.

Students Students practice creating melodies by sequencing note blocks. They learn to read simple musical notation and translate it to Scratch blocks.

Elaborate (10 min)

Teacher Program a sprite to play the KREIS school anthem or a familiar folk song. Write out the note sequence on paper first. Translate note letters to Scratch note numbers. Add appropriate timing with beat durations. Sync animation with the music by adding costume changes on beat.

Students Students create complete musical performances combining audio and visual elements. They practice planning before programming.

Evaluate (5 min)

Teacher Create a program that plays a five-note ascending scale. Add a second program that plays a descending scale. Combine both into a complete up-and-down scale. Explain how changing the tempo value affects the music playback speed.

Students Students demonstrate musical programming competence. They understand the relationship between tempo and playback speed.

Cross-Curricular Connections

Mathematics: Calculate the frequency in hertz for middle C which is note sixty using the formula four hundred forty times two raised to the power of note minus sixty-nine divided by twelve.
Science: Explain how digital audio converts musical notes into numerical values that computers can store and reproduce through speakers.
Language: Write a music theory connection explaining how mathematical ratios between notes create harmonious intervals.

Digital Citizen Reminder: Music compositions created in Scratch are original works. Share your musical creations with proper attribution. Do not copy copyrighted melodies without permission for public sharing.

Resources Needed:
- Scratch software with music extension
- Note frequency chart
- Song notation sheets
- Tempo reference
- Textbook

Differentiation

Approaching: Provide pre-written note sequences where students arrange them to create simple melodies
On Level: Students create standard melodies following musical notation guidelines
Advanced: Advanced students explore harmony by playing multiple notes simultaneously and create chord progressions
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Melody creation and music programming exercise
CT Skill Assessed: Sequential composition arranging elements in specific order to create meaningful output

Dormitory — Retrieval (Paper)

Hum a song you know and try to identify the notes. Are they going up or down? Write the pattern as up and down arrows.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

—

`( . . ) => . .`

Engage (5 min)

> **Teacher** Think about your favorite game. What makes it fun? Is it the challenge of avoiding obstacles, the satisfaction of collecting items, or the excitement of reaching new levels? What rules govern how the game works?

> **Students** Students analyze game elements and identify mechanics like scoring, collision, and level progression. They want to create games with similar engaging mechanics.

Explore (10 min)

> **Teacher** Play this simple Scratch game. Notice how the score increases when collecting coins. Lives decrease when touching enemies. The game ends when lives reach zero. Identify the three core mechanics: scoring, health, and game over condition.

> **Students** Students experience game mechanics from a player perspective. They identify how variables, conditions, and loops create the game experience.

Explain (10 min)

> **Teacher** Game mechanics are rules that control gameplay. Scoring uses a variable that changes based on player actions. Lives or health uses a variable that decreases when the player fails. Collision detection uses touching blocks to check if sprites interact. Game states include start screen, active gameplay, pause, and game over. The game loop uses a forever loop that checks conditions and updates the game each frame. Balancing mechanics means making the game challenging but not impossible.

> **Students** Students design game mechanics on paper before programming. They plan scoring rules, life systems, and win conditions.

Elaborate (10 min)

> **Teacher** Create a simple collection game. The player moves a character with arrow keys. Coins spawn randomly and the player must collect them. Each coin adds ten to score. An enemy moves back and forth and touching it costs one life. Three lives total and game ends when all lives are lost.

Students Students implement complete game mechanics. They practice combining movement, collision, scoring, and game state management.

Evaluate (5 min)

Teacher Test your game with a classmate. Observe whether the difficulty feels balanced. Is scoring too easy or too hard? Are lives fair? Adjust your game parameters and explain each change you made to improve the experience.

Students Students learn game testing and balancing. They understand that good game design requires iteration based on player feedback.

Cross-Curricular Connections

Mathematics: Calculate the average score a player achieves over ten game plays. Determine the standard deviation to measure score consistency.
Science: Explain how collision detection algorithms check bounding box overlap between sprites at each frame update.
Language: Write a game design document describing the rules, mechanics, and player experience for your game.

Digital Citizen Reminder: Games should be fair and inclusive. Avoid mechanics that frustrate specific player groups. Consider accessibility options like adjustable speed or difficulty settings.

Resources Needed:
- Scratch software
- Game design templates
- Balancing worksheet
- Player feedback forms
- Textbook

Differentiation

Approaching: Provide game templates with pre-built mechanics where students modify parameters and add features
On Level: Students create standard games following design guidelines
Advanced: Advanced students create power-up systems, multiple levels, and adaptive difficulty mechanics
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Game design document and playable prototype evaluation
CT Skill Assessed: Systems design creating interconnected mechanics that produce engaging emergent behavior

Dormitory — Retrieval (Paper)

Play a mobile game for five minutes tonight. Count how many different game mechanics you notice. Write down scoring, lives, power-ups, and obstacles.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

—

`(..) => ..`

Engage (5 min)

Teacher Think about how your phone recognizes your face to unlock. How does it know it is you and not someone else? Has the phone met you before? How did it learn to recognize your face among billions of faces?

Students Students wonder how devices learn to recognize individuals. They realize AI must learn from examples rather than being explicitly programmed for every possibility.

Explore (10 min)

Teacher Look at the Scratch ML extensions. Try the text recognition extension by typing hello and seeing how Scratch classifies it. Try the face detection extension to see how it tracks face position. Notice that Scratch can identify objects and detect motion.

Students Students experience machine learning through Scratch extensions. They see that AI can recognize text, faces, and objects from training data.

Explain (10 min)

Teacher Artificial intelligence or AI is computer software that mimics human thinking. Machine learning or ML is a type of AI that learns from data instead of following explicit rules. Training means showing the computer many examples so it learns patterns. Classification means assigning categories to new inputs based on training. A simple ML model takes input data, compares it to trained patterns, and outputs a prediction. AI powers voice assistants, recommendation systems, and translation tools.

Students Students discuss where AI appears in daily life. They list examples like voice assistants, photo tagging, and spam filters.

Elaborate (10 min)

Teacher Create a simple AI simulation in Scratch. Make a sprite that asks yes or no questions. Based on answers it tries to guess what the player is thinking. Use a decision tree of if-then-else blocks to narrow down possibilities. This simulates how simple AI makes decisions based on input.

Students Students build rule-based systems that simulate AI reasoning. They understand that simple AI follows decision trees based on user input.

Evaluate (5 min)

Teacher Explain the difference between AI and regular programs. Describe what training means in machine learning. Give two examples of AI you use regularly. Identify one limitation of current AI systems.

Students Students demonstrate understanding of AI concepts. They can distinguish between programmed rules and learned patterns.

Cross-Curricular Connections

Mathematics: Calculate the number of training examples needed if an AI needs one hundred examples per category and there are five categories to learn.
Science: Explain how neural networks inspired by human brain structure use mathematical weights to represent learned patterns.
Language: Write an argumentative essay about whether AI will help or harm education in Karnataka schools over the next ten years.

Digital Citizen Reminder: AI systems can reflect biases present in their training data. Always question whether AI recommendations are fair and accurate. Never blindly trust AI-generated content without verification.

Resources Needed:
- Scratch with ML extensions
- AI examples
- Machine learning visual demos
- Ethics discussion guide
- Textbook

Differentiation

Approaching: Provide simplified AI concept cards with pictures and examples for visual learners
On Level: Students complete standard AI identification and basic Scratch ML extension exercises
Advanced: Advanced students explore how training data quality affects AI accuracy and discuss bias in algorithms
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: AI concept quiz and Scratch ML extension exploration report
CT Skill Assessed: Conceptual understanding grasping how learning systems differ from traditional programs

Dormitory — Retrieval (Paper)

List three ways AI helps you on your phone. Think about what data the AI might have needed to learn to help you.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

—

`(..) => ..`

Engage (5 min)

Teacher You have learned variables, conditions, loops, lists, clones, pen drawing, music, and AI. Now you will combine everything into one capstone project. What kind of game would you love to create that uses as many of these skills as possible?

Students Students brainstorm ambitious game ideas that showcase their complete skill set. They feel excited to create something they can be proud of.

Explore (10 min)

Teacher Examine these capstone game examples from previous students. One game uses clones for enemies, variables for scoring, and loops for gameplay. Another uses pen for drawing paths and lists for level data. Identify which Scratch concepts each game uses.

Students Students analyze example projects and see how all concepts combine. They plan which concepts to include in their own games.

Explain (10 min)

Teacher A capstone project demonstrates your complete learning. Steps include brainstorming game concept, designing mechanics on paper, creating a project plan with features list, building the game in stages starting with core mechanics, adding polish with sounds and visuals, testing with classmates and fixing bugs, documenting your code with comments, and presenting with live demonstration.

Students Students create project plans with timelines and feature lists. They learn to break large projects into manageable tasks.

Elaborate (10 min)

Teacher Build your capstone game over this session and the next. Start with the core mechanic. Add scoring with variables. Include enemies or obstacles with clones. Create multiple levels with lists. Add sound effects and music. Include a start screen and game over screen. Test and balance the difficulty.

Students Students build complete games applying all learned concepts. They practice project management and iterative development.

Evaluate (5 min)

Teacher Present your capstone game to the class. Demonstrate gameplay live. Explain three programming concepts you used and how they contribute to the game. Answer questions from classmates and teacher about your design choices.

Students Students present their work professionally. They articulate technical decisions and respond to questions about their creative process.

Cross-Curricular Connections

Mathematics: Calculate the total development time for your game by tracking hours spent on planning, coding, testing, and documentation.
Science: Explain how game design applies psychological principles of motivation, reward, and challenge to keep players engaged.
Language: Write a game review analyzing another classmates game using criteria like fun factor, technical skill, and creative design.

Digital Citizen Reminder: Share your capstone game on Scratch with appropriate credits for any assets used. Respect other creators by not claiming their art or music as your own. Be proud of your original work.

Resources Needed:

- Scratch software
- All previous lesson materials
- Project planning templates
- Testing feedback forms
- Presentation area

Differentiation

Approaching: Provide game templates with core mechanics pre-built where students add features and polish
On Level: Students create complete games from scratch following project planning guidelines
Advanced: Advanced students create multi-level games with save systems, leaderboards, and adaptive difficulty
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Capstone project rubric with technical, creative, and presentation criteria
CT Skill Assessed: Synthesis integrating multiple skills and concepts into a cohesive interactive project

Dormitory — Retrieval (Paper)

Write a one-page game design document for your dream game. Describe the goal, rules, and what makes it fun. Think about what Scratch blocks you would need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Chapter 6 — Digital Citizenship

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Engage (5 min)

Teacher When you create a gaming account or social media profile you choose a name, a picture, and describe yourself. Is this the same person as the real you sitting in this classroom? How are your online and offline identities different?

Students Students discuss how they present themselves differently online and offline. They realize online identity is a constructed version of themselves.

Explore (10 min)

Teacher Look at these five social media profiles. Some use real names and photos. Others use fake names and cartoon avatars. Some share personal details while others share nothing. Which profiles seem trustworthy? Which seem suspicious? Why?

Students Students evaluate online profiles for trustworthiness. They notice that profiles with real photos and complete information seem more genuine.

Explain (10 min)

Teacher Online identity is how you present yourself on the Internet. It includes your username, profile picture, bio, and content you share. Your online identity can be the same as your real identity or completely different. Important considerations include privacy by limiting personal information, consistency by being the same person online and offline, reputation by creating content you would be proud of, and safety by protecting details like address and phone number.

Students Students analyze their own online identities. They evaluate what information they currently share and what they should protect.

Elaborate (10 min)

Teacher Design your ideal online identity for an educational platform. Choose an appropriate username that does not reveal your real name. Write a bio that shares your interests without personal details. Select or create a profile image that represents you without showing your face. Explain each choice and why it is safe.

Students Students create safe online identities. They practice sharing personality without compromising privacy.

Evaluate (5 min)

Teacher Evaluate these online identities for safety and appropriateness. Identify three pieces of information that should never be in an online profile. Create your own safe profile following all the guidelines we learned.

Students Students demonstrate understanding of online identity safety. They can identify risky information and create appropriate profiles.

Cross-Curricular Connections

Mathematics: Calculate how many different online accounts an average person has if they use separate accounts for gaming, social media, email, and school work.
Science: Explain how digital footprint is created when online activities are recorded on servers and can be traced back

to your identity.
Language: Write a personal online identity policy describing what information you will and will not share online.

Digital Citizen Reminder: Your online identity reflects on you permanently. Think before you post. What seems funny today might be embarrassing tomorrow or when applying for jobs in the future.

Resources Needed:
- Social media profile examples
- Online identity checklist
- Privacy settings guide
- Template profile forms
- Textbook

Differentiation

Approaching: Provide profile creation templates with safety guidelines embedded in each field
On Level: Students create standard online profiles following privacy and safety guidelines
Advanced: Advanced students research digital reputation management and create a comprehensive online presence strategy
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Online identity creation exercise with safety evaluation
CT Skill Assessed: Self-awareness understanding how personal information is perceived and used online

Dormitory — Retrieval (Paper)

Check the privacy settings on your phone apps tonight. Which apps have access to your camera, location, and contacts? Write down any settings you should change.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

> **Teacher** Your favorite game app asks for permission to access your contacts, location, camera, and microphone. Why does a simple game need all this information? What could it do with your contact list and location data?

Students Students realize that apps collect more data than needed for their function. They wonder why games need access to contacts and location.

Explore (10 min)

Teacher Open the settings on your phone. Look at the permissions for three different apps. Note what each app can access. Compare a camera app that needs camera access with a game that asks for camera access. Which permissions seem necessary and which seem suspicious?

Students Students discover that many apps request unnecessary permissions. They learn to evaluate whether permissions match app functionality.

Explain (10 min)

Teacher Personal data includes your name, age, address, phone number, email, photos, location, contacts, and browsing history. Apps collect this data for targeted advertising, service improvement, and sometimes selling to third parties. Privacy settings let you control what data apps can access. Data protection principles include collecting only necessary data, securing data with passwords, not sharing data without consent, and deleting data when no longer needed.

Students Students audit their own app permissions. They identify data types that need protection and learn to adjust privacy settings.

Elaborate (10 min)

Teacher Conduct a privacy audit of five apps on a school phone. For each app identify what data it collects, whether that data collection seems necessary, what privacy risks exist, and what settings could be changed to improve privacy. Write a report with recommendations.

Students Students practice real-world privacy analysis. They develop critical evaluation skills for digital privacy.

Evaluate (5 min)

Teacher Classify each data type as sensitive or non-sensitive. Phone number, favorite color, home address, school name, age. Explain why some data needs more protection than others. Create a list of five privacy rules everyone should follow.

Students Students demonstrate understanding of data sensitivity levels. They create practical privacy guidelines.

Cross-Curricular Connections

Mathematics: Calculate the percentage of apps on a phone that have unnecessary permissions if twelve out of twenty apps request more access than they need.
Science: Explain how data encryption protects personal information by converting readable text into scrambled code that only authorized parties can decode.
Language: Write a privacy guide for younger students explaining in simple terms why protecting personal data online is important.

Digital Citizen Reminder: Personal data is valuable. Companies collect it to make money through advertising. You have the right to know what data is collected about you and to request its deletion.

Resources Needed:
- Phone with app settings
- Privacy audit checklist
- Data type classification guide
- Permission review template
- Textbook

Differentiation

Approaching: Provide simplified permission review cards with pictures showing what each permission allows
On Level: Students complete standard privacy audit exercises on provided devices
Advanced: Advanced students research GDPR and Indian data protection laws and their impact on student privacy
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Privacy audit report and data classification exercise
CT Skill Assessed: Critical evaluation assessing digital tools for privacy implications and risks

Dormitory — Retrieval (Paper)

Tonight review the permissions for your three most-used apps. If any seem unnecessary, consider turning them off. Write down what changes you made.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

> **Teacher** Imagine someone creates a fake social media account using your photo and posts embarrassing messages pretending to be you. Your classmates see it and start sharing it. How would you feel? What would you do?

> **Students** Students empathize with the victim and begin discussing responses. They realize cyberbullying can happen to anyone and has serious emotional impact.

Explore (10 min)

Teacher Read these five scenarios on the board. One student receives threatening messages. Another has their photo edited without permission. A third is excluded from a group chat deliberately. A fourth has rumors spread about them online. A fifth is mocked for their answers in class group. Identify which scenarios are cyberbullying and explain your reasoning.

Students Students analyze scenarios and identify cyberbullying behaviors. They recognize that exclusion and mocking are also forms of bullying.

Explain (10 min)

Teacher Cyberbullying is using technology to hurt, embarrass, or threaten another person repeatedly. Forms include sending threatening messages, spreading rumors online, sharing embarrassing photos without consent, creating fake profiles to humiliate someone, deliberately excluding someone from online groups, and impersonating someone to damage their reputation. Impact includes anxiety, depression, loneliness, and decreased academic performance. Prevention strategies include not responding to bullies, blocking and reporting them, saving evidence, telling a trusted adult, and supporting victims.

Students Students learn to identify and respond to cyberbullying. They practice response strategies through role-play scenarios.

Elaborate (10 min)

Teacher Create a cyberbullying prevention poster for the school. Include the definition of cyberbullying, five warning signs that someone is being cyberbullying, steps to take if you are being bullied, how to support a friend who is being bullied, and helpline numbers. Use clear language appropriate for all ages.

Students Students create educational materials for their school community. They apply their knowledge to help others.

Evaluate (5 min)

Teacher Respond to these situations. You receive a mean message from an unknown number. Your friend shows you a cruel meme about another classmate. You accidentally see hurtful comments on a classmates post. Explain the appropriate response for each situation.

Students Students demonstrate appropriate responses to cyberbullying scenarios. They show understanding of when to report and when to intervene.

Cross-Curricular Connections

Mathematics: Calculate the estimated number of students affected by cyberbullying if research shows thirty percent of students experience it and your school has five hundred students.
Science: Explain how cyberbullying affects brain development and mental health through stress hormones like cortisol and adrenaline.
Language: Write a persuasive speech about why cyberbullying should be taken as seriously as physical bullying in school policies.

Digital Citizen Reminder: Being a bystander who does nothing makes you part of the problem. Always report cyberbullying when you see it. Support victims and stand up against bullying even when it is difficult.

Resources Needed:

- Cyberbullying scenario cards
- Prevention poster materials
- Helpline directory
- Role-play scripts
- Textbook

Differentiation

Approaching: Provide simplified scenario cards with clear illustrations showing bullying and non-bullying behaviors
On Level: Students analyze standard cyberbullying scenarios and practice response strategies
Advanced: Advanced students research cyberbullying laws and create a school policy proposal for digital behavior
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Cyberbullying scenario response evaluation and poster creation
CT Skill Assessed: Empathetic response applying ethical principles to protect others in digital spaces

Dormitory — Retrieval (Paper)

Think about how you treat others in group chats and gaming. Have you ever said something online that you would not say face to face? What should you do differently?

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Everything you post online stays online forever. Even if you delete a photo someone may have already screenshot it. Imagine a future employer searching your name online. What would they find? Would you be proud of what they see?

Students Students realize that online actions have permanent consequences. They begin to think about their digital legacy.

Explore (10 min)

Teacher Search for your own name on the Internet. What appears? Now search for the name of a famous person and see how many years of content appears. Discuss how a digital footprint builds over time and how difficult it is to remove content once it is online.

Students Students discover their own digital footprint. They see that public figures have extensive digital histories they cannot control.

Explain (10 min)

Teacher A digital footprint is the trail of data you leave when using the Internet. Active footprints are things you intentionally post like social media updates and comments. Passive footprints are data collected without your direct action like browsing history, location data, and search queries. Companies store this data on servers indefinitely. Future consequences include college admissions, job applications, and personal relationships being affected by past online behavior. Managing your footprint means thinking before posting, using privacy settings, and regularly reviewing your online presence.

Students Students map their own digital footprints. They categorize activities into active and passive footprints.

Elaborate (10 min)

Teacher Create a digital footprint timeline of your life so far. Mark when you first went online, created your first account, made your first post, and other milestones. Estimate what data exists about you online. Write a letter to your future self about what digital footprint you want to leave.

Students Students reflect on their digital history and plan their digital future. They understand that they are actively creating their permanent record.

Evaluate (5 min)

Teacher Define this person's digital footprint. What can you learn about them from their online presence? Is there anything concerning? How could they improve their digital footprint? Apply the same analysis to your own online presence.

Students Students develop critical awareness of digital footprint implications. They can evaluate online presence from an outsider perspective.

Cross-Curricular Connections

Mathematics: Calculate how many gigabytes of data a person might generate annually if they post five photos per day at three megabytes each plus daily social media activity.
Science: Explain how web servers use IP addresses and cookies to track user behavior across different websites building comprehensive profiles.
Language: Write a reflective essay about your digital footprint titled What I Want My Online Legacy to Be.

Digital Citizen Reminder: Your digital footprint is your permanent reputation. Before posting ask yourself if you would be comfortable with your teacher, parents, or future employer seeing this content.

Resources Needed:
- Internet search demonstration
- Digital footprint timeline template
- Privacy settings guide

- Reflection journal prompts
- Textbook

Differentiation

Approaching: Provide simplified digital footprint tracking sheets with icons representing common online activities
On Level: Students complete standard digital footprint analysis exercises
Advanced: Advanced students research data brokers and explore tools for removing personal information from commercial databases
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Digital footprint analysis and management action plan
CT Skill Assessed: Long-term thinking evaluating how current actions affect future outcomes

Dormitory — Retrieval (Paper)

Search for your name online tonight. If nothing appears that is good. If something appears think about whether you would want a future teacher to see it.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

> **Teacher** Two students apply for the same college scholarship. The admission officer searches both names online. Student A has a blog about science projects and coding achievements. Student B has inappropriate photos and complaints from friends. Who gets the scholarship?

> **Students** students immediately recognize that online reputation affects real-world opportunities. They understand that digital presence influences how others perceive them.

Explore (10 min)

> **Teacher** Examine these three online profiles. Profile one belongs to a student who posts only complaints and negative comments. Profile two belongs to a student who shares school projects and volunteer work. Profile three belongs to a student who has no online presence at all. Which profile would impress a teacher or employer most?

Students students evaluate how different online behaviors create different reputations. They realize that having no presence is also a choice with consequences.

Explain (10 min)

Teacher Online reputation is how others perceive you based on your digital presence. It affects college admissions, job opportunities, friendships, and personal relationships. Building positive reputation requires sharing achievements and interests, commenting respectfully, connecting with mentors and role models, creating original content that shows your skills, and being consistent in positive behavior. Recovery from reputation damage involves removing harmful content, creating positive content to push down negative results, apologizing sincerely if you wronged others, and building new positive history over time.

Students students assess their own online reputation. They create an action plan for building positive digital presence.

Elaborate (10 min)

Teacher Audit your current online reputation. Search your name and document what appears. Identify any content that could damage your reputation. Create a plan to improve your online presence by listing five positive actions you can take this month. Start implementing one action today.

Students students take concrete steps to improve their digital reputation. They practice proactive reputation management.

Evaluate (5 min)

Teacher Evaluate these online behaviors for reputation impact. Posting homework help requests. Sharing science fair project results. Leaving negative reviews of restaurants. Commenting on friends photos respectfully. Arguing with strangers online. Rate each as positive or negative for reputation.

Students students demonstrate understanding of how specific online actions affect reputation. They can distinguish between reputation building and damaging behaviors.

Cross-Curricular Connections

Mathematics: Calculate how many positive blog posts would be needed to push a negative article off the first page of search results assuming each result is equally likely to be clicked.
Science: Explain how search engine algorithms rank results based on relevance, recency, and authority of content about a person.
Language: Write a personal branding strategy document describing how you want to be perceived online and the actions needed to achieve that perception.

Digital Citizen Reminder: Online reputation is earned through consistent positive behavior not through deleting negative content. Focus on creating value for others rather than managing perception superficially.

Resources Needed:
- Reputation audit template
- Search engine demonstration
- Brand building guide
- Portfolio examples
- Textbook

Differentiation

Approaching: Provide reputation audit worksheets with guided questions for analyzing search results
On Level: Students complete standard reputation analysis and improvement planning exercises
Advanced: Advanced students create professional online portfolios showcasing their skills and achievements
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Reputation audit report and improvement action plan
CT Skill Assessed: Strategic thinking planning long-term reputation management through consistent positive action

Dormitory — Retrieval (Paper)

Ask a family member what they would think if they searched your name online. Write down their honest reaction and consider what changes you might want to make.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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Engage (5 min)

Teacher Your friend writes a poem and shares it online. Someone else copies it, puts their name on it, and submits it as their own work. Is this legal? Does your friend have any rights over the poem they created?

Students students recognize that creators have rights over their original work. They begin to understand the concept of intellectual property.

Explore (10 min)

Teacher Look at these items. A published textbook. A movie. A song. A student essay. A recipe. A mathematical formula. A news photograph. Which of these are protected by copyright? Which are not? Why are some protected and others not?

Students students identify that creative works are protected while facts and ideas are not. They understand that copyright applies to original expression.

Explain (10 min)

Teacher Copyright is a legal right that gives creators control over their original works. It automatically applies when a work is created in a fixed form. Copyright protects books, music, movies, software, photographs, and artwork. It does not protect ideas, facts, or methods. Copyright gives the creator exclusive rights to reproduce, distribute, perform, and display their work. Duration varies but generally lasts for the creator lifetime plus sixty years. Works enter public domain when copyright expires and can be freely used.

Students students learn copyright basics. They identify which of their own schoolwork is protected by copyright automatically.

Elaborate (10 min)

Teacher Create a copyright awareness guide for KREIS students. Include the definition of copyright, examples of protected works, what you can and cannot do with copyrighted material, how to find copyright-free resources, and what fair use means for school projects. Use clear examples relevant to student life.

Students students create practical copyright guides for their peers. They apply copyright knowledge to educational context.

Evaluate (5 min)

Teacher Is this copyright infringement. Copying a chapter from a textbook for your own use. Using a movie clip in your school presentation. Selling copies of a song you downloaded. Quoting a paragraph in your essay with credit. Explain why each is or is not infringement.

Students students demonstrate understanding of copyright application. They can distinguish between infringement and permitted use.

Cross-Curricular Connections

Mathematics: Calculate how long copyright lasts for a book if the author died in 1960. Add the author lifetime plus sixty years to determine when it enters public domain.
Science: Explain how digital technology makes copyright enforcement challenging because perfect copies can be made and distributed instantly.
Language: Write a letter to a classmate explaining why it is wrong to copy someone else's homework using copyright concepts.

Digital Citizen Reminder: Respect copyright by always crediting original creators. Use legitimate sources for your projects. Remember that not being caught does not make infringement acceptable.

Resources Needed:
- Copyright examples
- Public domain resources
- Fair use guidelines
- Creative Commons introduction
- Textbook

Differentiation

Approaching: Provide copyright scenario cards with clear yes and no examples for basic understanding
On Level: Students complete standard copyright identification and fair use analysis exercises
Advanced: Advanced students explore international copyright differences and digital rights management technologies
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Copyright application quiz with scenario analysis
CT Skill Assessed: Legal reasoning applying intellectual property rules to real-world situations

Dormitory — Retrieval (Paper)

Look at the textbook you use every day. Who owns the copyright? Find the copyright notice on the first few pages and write down the year and owner name.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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(..) => ..
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Engage (5 min)

Teacher Copyright says you cannot use others work without permission. But what if creators want to share their work freely while still getting credit? Is there a way to give permission in advance to everyone?

Students students realize that copyright is restrictive and wonder if there are alternatives for sharing. They discover that Creative Commons provides flexible licensing.

Explore (10 min)

Teacher Visit the Creative Commons website. Look at the six main license types. Match each symbol to its meaning. The BY symbol means attribution required. The NC symbol means non-commercial only. The ND symbol means no derivatives. The SA symbol means share alike. Find images on a CC search engine using different license filters.

Students students explore Creative Commons licenses visually. They practice searching for content with specific license requirements.

Explain (10 min)

Teacher Creative Commons or CC provides free licenses that let creators share work with specific permissions. CC BY requires giving credit to the creator. CC BY-SA requires credit and sharing under the same license. CC BY-NC requires credit and non-commercial use only. CC BY-ND requires credit and no modifications allowed. CC0 or public domain dedication means no restrictions at all. When using CC content always follow the license requirements. Attribution should include creator name, title, source, and license type.

Students students practice creating proper attribution for CC content. They learn to match their project needs with appropriate license types.

Elaborate (10 min)

Teacher Find three Creative Commons images for a school presentation on Karnataka culture. Write proper attribution for each image following CC guidelines. Explain why you chose each image and how its license affects what you can do with it. Include the attributions in your presentation credits.

Students students practice finding and using CC content legally. They build attribution habits for ethical content use.

Evaluate (5 min)

Teacher Match each Creative Commons license to its requirements. Identify which licenses allow commercial use. Explain what share alike means. Create proper attribution for this CC BY-NC-SA image.

Students students demonstrate CC license knowledge. They can apply licensing rules correctly in their projects.

Cross-Curricular Connections

Mathematics: Calculate the number of possible license combinations when there are four license elements that can be included or excluded independently.
Science: Explain how Creative Commons licenses work legally by giving advance permission while maintaining creator control over certain conditions.
Language: Write a comparison of copyright, Creative Commons, and public domain explaining when to use each.

Digital Citizen Reminder: Even when content is free to use under CC licenses, always give proper attribution. Failing to credit creators 违反 s the license and disrespects their generosity in sharing.

Resources Needed:
- Creative Commons website
- License symbol reference
- CC search engine
- Attribution template
- Textbook

Differentiation

Approaching: Provide CC license matching cards with simplified explanations for each license type
On Level: Students complete standard CC license identification and attribution exercises
Advanced: Advanced students explore CC0 public domain dedication and compare CC with other open content licenses
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: CC license identification and attribution creation exercise
CT Skill Assessed: Ethical application applying licensing rules to respect creators while enabling sharing

Dormitory — Retrieval (Paper)

Search for a Creative Commons image about your favorite topic. Save the image and write the complete attribution including creator, title, source, and license.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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`(..) => ..`

Engage (5 min)

Teacher You see a news article on social media that says eating bananas cures all diseases. It has thousands of shares and comments. Does this make it true? How can you tell if news is real or fake before sharing it with others?

Students students realize that popularity does not equal truth. They want to learn how to verify information before believing and sharing.

Explore (10 min)

Teacher Examine these four news items. One is from a recognized newspaper with named sources. One is from an anonymous blog with no sources. One exaggerates findings with sensational language. One presents balanced facts with multiple viewpoints. Rate each for credibility and explain your ratings.

Students students identify credibility indicators like named sources, balanced reporting, and professional presentation versus anonymous sources and sensational language.

Explain (10 min)

Teacher Fake news is false information presented as real news. Types include propaganda with deliberate political bias, satirical content mistaken as real, misleading content with exaggerated headlines, manipulated photos or videos, and fabricated stories with no factual basis. Verification techniques include checking the source reputation, looking for named experts and sources, searching for the same story on reliable news sites, examining the date to avoid old news, checking photos using reverse image search, and reading beyond the headline to the full article.

Students students practice verification techniques on sample stories. They learn to pause and verify before sharing information.

Elaborate (10 min)

Teacher Create a fake news detection checklist for KREIS students. Include at least ten verification steps. Test your checklist by evaluating three news stories two real and one fake. Document your verification process for each story and explain your final judgment.

Students students build practical verification tools. They apply systematic checking to real information evaluation.

Evaluate (5 min)

Teacher Evaluate this news story. Check the source. Look for named experts. Search for corroborating reports. Examine the language for bias. Check the date. Write a credibility report explaining whether the story appears true or false and why.

Students students apply comprehensive verification methodology. They can support their credibility judgments with specific evidence.

Cross-Curricular Connections

Mathematics: Calculate the percentage of people who share news articles without reading beyond the headline based on research showing sixty percent of shares happen without clicking the link.
Science: Explain how social media algorithms create filter bubbles that show users only content matching their existing beliefs making them more susceptible to fake news.
Language: Write a media literacy guide for classmates explaining five steps to verify any news story before sharing it.

Digital Citizen Reminder: Sharing fake news harms society by spreading misinformation. Always verify before sharing. You are responsible for the information you spread even if you did not create it.

Resources Needed:

- Fake news examples
- Verification tools guide
- Source evaluation rubric
- Reverse image search demo
- Textbook

Differentiation

Approaching: Provide simplified verification checklists with visual cues for identifying fake news indicators
On Level: Students complete standard source evaluation and fact-checking exercises
Advanced: Advanced students explore deepfake technology and analyze how AI-generated fake content can be detected
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Fake news detection exercise with verification documentation
CT Skill Assessed: Critical analysis evaluating information credibility through systematic verification

Dormitory — Retrieval (Paper)

Look at the news on your phone tonight. Pick one story and check if the same story appears on other reliable news websites. Write whether you found confirmation.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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`(..) => ..`

Engage (5 min)

> **Teacher** How many hours did you spend looking at screens yesterday including phone, computer, and television? Research suggests teenagers should have no more than two hours of recreational screen time daily. Was your screen time above or below this recommendation?

> **Students** students estimate their screen time and realize they may exceed recommended limits. They want to understand the health effects.

Explore (10 min)

> **Teacher** Keep a screen time diary for three days. Record every screen activity with start time and end time. Calculate total daily screen time. Categorize activities as educational, social, entertainment, or creative. What percentage of your screen time is educational versus entertainment?

> **Students** students collect real data about their own screen habits. They discover patterns in how they spend their screen time.

Explain (10 min)

> **Teacher** Excessive screen time can cause eye strain, headaches, poor posture, sleep disruption from blue light, reduced physical activity, and decreased attention span. Balance strategies include setting daily time limits, taking breaks every thirty minutes using the twenty-twenty-twenty rule of looking at something twenty feet away for twenty seconds, creating screen-free zones like bedrooms and dining areas, replacing some screen time with physical activities, and scheduling specific times for recreational screen use.

> **Students** students evaluate their own screen habits against health guidelines. They identify areas where they can reduce screen time.

Elaborate (10 min)

> **Teacher** Design a balanced daily schedule for a KREIS school day that includes study time with screens, recreational screen time limited to two hours, physical activities, social time with friends without screens, and sleep preparation without screens for one hour before bed. Explain the health reasoning behind each time allocation.

Students students create practical schedules that balance technology use with healthy habits. They apply health knowledge to personal planning.

Evaluate (5 min)

Teacher Rate your current screen habits as healthy or unhealthy for each category. Eye care. Sleep habits. Physical activity. Social interaction. Academic focus. Create an action plan to improve your weakest area.

Students students self-assess their screen habits honestly. They create specific actionable improvement plans.

Cross-Curricular Connections

Mathematics: Calculate total weekly screen time if you average three hours daily on weekdays and five hours on weekends.
Science: Explain how blue light from screens suppresses melatonin production affecting sleep quality and circadian rhythm.
Language: Write a personal wellness pledge describing three specific screen time balance commitments you will follow.

Digital Citizen Reminder: Technology is a tool that should serve you not control you. Be intentional about your screen time. Choose quality content over endless scrolling. Your attention is valuable.

Resources Needed:
- Screen time tracking app
- Health guidelines poster
- Balance schedule template
- Break timer apps
- Textbook

Differentiation

Approaching: Provide simplified screen time tracking sheets with picture categories for different types of screen use
On Level: Students complete standard screen time analysis and balance planning exercises
Advanced: Advanced students research digital wellness apps and present strategies for managing technology addiction
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Screen time diary analysis and balance plan creation
CT Skill Assessed: Self-regulation monitoring and modifying personal technology use for health

Dormitory — Retrieval (Paper)

Tonight set a timer for your screen use. When the timer goes off put your device down and do something else. Write down how long you actually used the screen.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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`(..) => ..`

Engage (5 min)

Teacher Have you ever felt anxious when your phone battery dies? Or felt stressed when you cannot check social media for an hour? Or stayed up late scrolling through videos unable to stop? These feelings might indicate technology is controlling you rather than you controlling it.

Students students recognize their own technology-related anxiety. They begin to understand unhealthy technology relationships.

Explore (10 min)

Teacher Complete this digital wellness self-assessment. Rate each statement from one to five. I feel anxious without my phone. I check social media first thing in the morning. I use screens to avoid uncomfortable feelings. I lose track of time when using devices. I have arguments about screen time. Calculate your total score and identify your wellness level.

Students students self-assess their digital wellness honestly. They identify areas where technology use has become problematic.

Explain (10 min)

Teacher Digital wellness means maintaining a healthy relationship with technology. Signs of technology addiction include feeling anxious without devices, checking phones compulsively, using screens to escape problems, neglecting face-to-face relationships, and experiencing sleep problems from late-night use. Healthy habits include setting phone-free times, practicing mindful technology use by being intentional about each interaction, taking digital detox periods regularly, engaging in offline hobbies, and maintaining real-world friendships through in-person activities.

Students students create personal digital wellness plans. They identify specific healthy habits to develop.

Elaborate (10 min)

Teacher Design a digital wellness challenge for your dormitory. Create seven daily challenges one for each day of the week that encourage healthier technology use. Examples include phone-free meals, thirty-minute outdoor time, reading a physical book, and having a face-to-face conversation without phones.

Students students create community wellness initiatives. They practice promoting healthy habits among peers.

Evaluate (5 min)

Teacher Identify three signs of unhealthy technology use in your own habits. Create specific strategies to address each sign. Plan a one-day digital detox and describe what you would do instead of screen time.

Students students recognize personal unhealthy patterns and create concrete improvement strategies. They plan practical detox experiences.

Cross-Curricular Connections

Mathematics: Calculate the number of hours per week spent on screens versus physical activities and compare against WHO recommendations for teenagers.
Science: Explain how dopamine release from social media notifications creates addiction patterns similar to other behavioral addictions.
Language: Write a reflective journal entry about your relationship with technology and what changes would improve your digital wellness.

Digital Citizen Reminder: Digital wellness is a personal responsibility. Help your friends recognize unhealthy technology habits. Support each other in building healthier digital lifestyles.

Resources Needed:

- Digital wellness assessment
- Detox challenge cards
- Mindfulness exercises
- Wellness plan template
- Textbook

Differentiation

Approaching: Provide simplified wellness checklists with visual scales for self-assessment
On Level: Students complete standard wellness analysis and planning exercises
Advanced: Advanced students research neuroscience of technology addiction and present findings to classmates
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Digital wellness assessment and personal improvement plan
CT Skill Assessed: Self-awareness recognizing personal patterns and taking responsibility for digital health

Dormitory — Retrieval (Paper)

Try a one-hour digital detox tonight. Put your phone in a drawer and do something else. Write down how you felt during that hour.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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`(..) => ..`

Engage (5 min)

Teacher A classmate posts a photo from a school trip. Someone comments This is so ugly and tags their friends to laugh. The original poster feels embarrassed and hurt. Was the comment necessary? Could the commenter have said nothing or been kind instead?

Students students recognize that online comments affect real people with real feelings. They understand that posting is a social act with consequences.

Explore (10 min)

Teacher Read these five online comments on a classmate's project post. One is encouraging. One is constructive criticism. One is sarcastic and mean. One is irrelevant spam. One shares the project with others complimenting the work. Rate each comment from positive to harmful and explain your rating.

Students students evaluate different types of online comments. They identify which contribute positively and which cause harm.

Explain (10 min)

Teacher Every online post affects others and your reputation. Before posting use the THINK framework. Is it True? Do I need to say it? Is it Helpful? Is it Inspiring? Is it Necessary? Is it Kind? Positive posting includes congratulating achievements, offering constructive feedback, sharing useful information, encouraging struggling classmates, and celebrating others successes. Avoid posting when angry, sharing others private information, making hurtful comments, or participating in pile-on criticism.

Students students practice applying the THINK framework to sample posts. They learn to pause and evaluate before publishing.

Elaborate (10 min)

Teacher Review your last ten social media posts or messages. Apply the THINK framework to each. Rate each as positive, neutral, or negative. Identify any posts you would delete or rewrite. Create a personal posting guide with rules you will follow for future posts.

Students students honestly evaluate their own posting habits. They create personal guidelines for responsible online communication.

Evaluate (5 min)

Teacher Evaluate these posts using the THINK framework. A sarcastic comment about a classmates mistake. A congratulatory message on a friends achievement. Sharing someone photo without asking permission. A detailed complaint about a teacher online. Rate each and explain the appropriate response.

Students students apply ethical reasoning to posting decisions. They demonstrate understanding of impact and responsibility.

Cross-Curricular Connections

Mathematics: Calculate the potential reach of a negative comment if it is shared by ten people who each share with ten more people creating exponential spread.
Science: Explain how neuroscience shows that receiving negative online comments activates the same brain areas as physical pain.
Language: Write a code of conduct for your class social media group outlining rules for respectful and responsible posting.

Digital Citizen Reminder: You are responsible for every word you post online. Kindness costs nothing but meanness can cause lasting damage. When in doubt do not post. Your words define who you are online.

Resources Needed:
- THINK framework poster
- Comment evaluation cards
- Personal posting guide template
- Positive posting examples
- Textbook

Differentiation

Approaching: Provide simplified THINK framework cards with visual prompts for each question
On Level: Students complete standard posting evaluation and personal guideline creation exercises
Advanced: Advanced students explore online community moderation and create guidelines for managing group discussions
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Posting evaluation exercise and personal posting guide creation
CT Skill Assessed: Ethical judgment applying values-based decision making to online communication

Dormitory — Retrieval (Paper)

Before you post anything on social media tonight, ask yourself each question of the THINK framework. Is it True? Helpful? Inspiring? Necessary? Kind? Write down whether you posted and why.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

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(..) => ..
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Engage (5 min)

Teacher Throughout this chapter we have learned about online identity, privacy, cyberbullying, digital footprints, reputation, copyright, fake news, screen time, digital wellness, and responsible posting. How can you make sure you remember and follow all these lessons every day?

Students students realize that knowledge without commitment is ineffective. They want to create personal pledges to guide their digital behavior.

Explore (10 min)

Teacher Examine these three digital citizenship pledges from students in other schools. One focuses on privacy. One focuses on kindness. One covers all major topics comprehensively. Which pledge seems most complete and practical? What topics would you add?

Students students evaluate existing pledges and identify gaps. They realize that comprehensive pledges cover multiple aspects of digital citizenship.

Explain (10 min)

Teacher A digital citizenship pledge is a personal commitment to behave responsibly and ethically online. It should include privacy commitments about protecting personal data and respecting others privacy, safety commitments about avoiding cyberbullying and reporting harmful behavior, integrity commitments about honest posting and respecting copyright, wellness commitments about balanced screen time and healthy habits, and community commitments about supporting others and building positive digital spaces.

Students students draft comprehensive pledges covering all topics learned. They personalize commitments to their specific situations.

Elaborate (10 min)

Teacher Write your complete digital citizenship pledge. Include at least ten specific commitments covering identity, privacy, bullying, footprint, reputation, copyright, fake news, screen time, wellness, and posting. Make each commitment specific and actionable. Design an appealing presentation for your pledge. Sign it and date it. Share it with your class.

Students students create formal pledges demonstrating their complete understanding. They commit to specific behaviors they will follow.

Evaluate (5 min)

Teacher Present your digital citizenship pledge to the class. Explain why you included each commitment. Receive feedback from classmates on completeness and specificity. Revise your pledge based on feedback. Create an action plan for how you will remind yourself of your commitments daily.

Students students present their pledges confidently. They demonstrate integrated understanding of all digital citizenship concepts.

Cross-Curricular Connections

Mathematics: Calculate how many commitments you make in your pledge and estimate the number of daily digital decisions you make that would be guided by each commitment.
Science: Explain how digital citizenship relates to civic responsibility and how being a good digital citizen contributes to a better society.
Language: Write a letter to the school principal proposing that digital citizenship be included in the school code of conduct with specific examples from your pledge.

Digital Citizen Reminder: A pledge is only meaningful if you follow through. Post your pledge where you will see it daily. Review it weekly. Update it as you grow and learn. Hold yourself accountable and ask friends to remind you of your commitments.

Resources Needed:
- Pledge examples
- Digital citizenship review guide
- Presentation materials
- Accountability partner system
- Textbook

Differentiation

Approaching: Provide pledge templates with sentence starters for each commitment area
On Level: Students create complete pledges covering all digital citizenship topics
Advanced: Advanced students create peer education materials teaching digital citizenship to younger students
SEND: Touch and identify network devices with teacher guidance, use simple one-word labels

Formative Assessment

Method: Digital citizenship pledge evaluation and peer review
CT Skill Assessed: Commitment and synthesis integrating all concepts into actionable personal principles

Dormitory — Retrieval (Paper)

Read your digital citizenship pledge tonight before sleeping. Choose one commitment to focus on tomorrow. Write down the specific action you will take to follow that commitment.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Colophon

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